

The Making of Dominant Vehicle Currencies: Evidence from DeFi

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Abstract

We reconstruct 475 million swaps and observe the vehicle inside each connected route. From 2024 H1 to 2026 H1, stablecoin share rises 25.5 percentage points (pp). Net switching within continuing pairs is -0.1 pp, while pairs first observed after 2024 H1 contribute $+20.1$ pp. The entry vehicle predicts subsequent use. At the first sampled exact contest, it carries 83.1% of routes and current output predicts retention. When exact-output leadership changes hands, route share follows; prior weak-leg capital predicts whether the new lead persists. Dominance therefore shifts mainly through pair formation and activity reallocation, while prices and two-leg depth sustain vehicles inside established pairs.

1 Introduction

Vehicle currencies can dominate through transitory use. Traders acquire them to exchange them again because two liquid markets through the vehicle can outperform a thin direct market. The U.S. dollar appears on more than 85% of foreign-exchange transactions, far more often than the United States' share of cross-border economic activity would suggest (Somogyi, 2026). Vehicle use concentrates trading in dollar markets; deeper markets lower trading costs and reinforce the dollar's intermediary role (Krugman, 1980; Kiyotaki and Wright, 1989; Flandreau and Jobst, 2009). Market-making capacity also determines the supply of immediacy (Grossman and Miller, 1988). Shared intermediaries can generate common liquidity across markets (Coughenour and Saad, 2004). This feedback explains persistence after dominance. We ask how a challenger gains share as new trading relationships and market technologies appear.

Aggregate dominance combines two margins. A continuing relationship can replace its vehicle, or activity can move toward relationships that already use another vehicle and toward new relationships formed around it. The same aggregate change can therefore arise with pervasive switching or with no net switching inside established relationships. Separating these cases requires transactions that attach the intermediary to the exchange between its source and destination.

Standard foreign-exchange records report the two legs of a customer's exchange separately.¹ A yen-euro exchange routed through dollars appears as separate yen-dollar and dollar-euro trades.

¹For example, see the [BIS 2025 Triennial Survey reporting guidelines and templates](#).

Somogyi (2026) infers vehicle activity from currency-pair volume responses around non-overlapping holidays. In decentralised exchange, the pool calls inside one blockchain transaction preserve their execution order. We use those records to observe the connected route directly.

We call the ordered source and destination the *endpoint pair*, shortened hereafter to *pair*. Each directed pool execution is a *leg*; the ordered sequence of connected legs is a *route*; and an asset between the two endpoints is a vehicle. We reconstruct routes from 475 million swaps on 9 Ethereum automated market maker (AMM) deployments from November 2018–June 2026; Table 1 reports the resulting panel. The token sequences identify the pair, every intermediary, and whether the route spans exchanges. The connected transaction identifies vehicle use directly.

The sample begins with an institutional benchmark for how a vehicle network forms. Uniswap v1 allowed only ETH–token pools, so each token-to-token trade had to cross ETH; we recover 217,003 such routes. Uniswap v2 removed that rule and allowed any two ERC-20 tokens to form a pool. Six years later, however, 95.5% of single-leg v2 trades still use a WETH pool and 97.9% of token combinations first traded in 2026 include WETH. Appendix Table 11, panels A and B, reports these institutional facts. The native-centred pool network survived removal of the mandate. We read this chronology descriptively because venue migration and available token combinations changed with the rule.

Stablecoins, including USDC, USDT, DAI, and other pegged tokens, become substantially more prominent vehicles later in the sample. Their exact two-leg route share rises from 16.9% to 42.4% between matched 2024 and 2026 dates. Of this +25.5 pp increase, net switching within continuing pairs contributes −0.1 pp, activity reallocation across continuing pairs contributes +8.4 pp, and pairs first observed after 2024 H1 contribute +20.1 pp before the remaining lifecycle margins enter (Table 2, panel A). The dollar-weighted decomposition in panel B has the same extensive-margin structure. Figure 1 also shows an earlier reversal toward native vehicles, tying the rotation to pair formation and the distribution of trading across pairs instead of a smooth technology trend.

Pair-level estimates trace how that new vehicle use becomes established. Among pairs that trade again, a 10 pp higher entry-day stablecoin share predicts +8.92 pp higher stablecoin share during days 1–30 and +8.40 pp during days 31–120 (Table 3, panel A, columns 2 and 5). After prior-calendar full-range capital reaches both stablecoin legs, 38.3% of pairs use the bridge within 30 days and 47.0% within 120 days (Table 4, panel A); relative weak-leg depth predicts subsequent route allocation (panel C). At the first sampled exact contest, the entry family still carries 83.1% of routes, while its exact-output advantage predicts retention and its separate prior-capital coefficient is imprecise (Table 5, panels A and B). Bridge support precedes adoption; exact output predicts retention when the vehicles first compete.

Competition becomes sharper inside established pairs. On the same route sample, a 100 bp incumbent output advantage predicts +10.13 pp higher retention, and a 10 pp prior full-range capital-share advantage predicts another +2.77 pp (Table 6, column 4). When exact-output leadership changes hands, incumbent share falls 29.0 pp more than during the same event’s earlier movement (Table 7, panel B, column 4). Prior challenger capital predicts whether the new price

lead persists (column 2), while routes retaining a mature incumbent leave only 3.3 bp of input-value-weighted gross output (Table 8, panel B). Gas changes the receipt-matched aggregate shortfall from 7.46 to 7.65 bp and has its larger effect below \$1,000 (panel D). Vehicle dominance is thus maintained through price competitiveness associated with two-sided depth, while exact-price rank reversals reallocate flow.

We contribute an extensive-margin account of currency dominance and evidence on the market structure that sustains an incumbent vehicle. The route panel complements the holiday-based inference of [Somogyi \(2026\)](#) and the historical network evidence of [Flandreau and Jobst \(2009\)](#). The pair decomposition connects aggregate leadership to relationship formation; the post-entry and exact-price estimates connect formation to classical volume–cost feedback, constrained currency-market liquidity supply, and decentralised-exchange liquidity provision ([Krugman, 1980](#); [Huang et al., 2025](#); [Lehar and Parlour, 2025](#)). The price comparison extends optimal-routing work from parallel pools to sequential routes through alternative vehicles ([Xi and Moallemi, 2026](#)). Together, the results show where a vehicle network expands, when a cheaper challenger takes flow, and how two-leg capital predicts the persistence of price leadership.

The extensive margin is also relevant for payment-system design. Project Nexus assigns currency conversion to a foreign-exchange provider, while Project Rialto uses a third currency when direct exchange is unavailable ([Bank for International Settlements Innovation Hub, 2024, 2025](#)). Access policy can therefore affect the price inside an existing relationship and which two-sided markets become the bridge inherited by new ones.

2 Setting and data

2.1 Routes and notation

Decentralised exchanges distribute liquidity across smart-contract pools that routers can combine within one transaction. A trader may use one pool, divide an order across several pools, or trade through an intermediary asset. Because the pool calls share a transaction, the intermediary sequence is observable.² An intermediary route supplies connectivity when a direct pool is absent; when direct and indirect pools coexist, the router compares their fees, price impact, gas, and venue access. [Lehar and Parlour \(2025\)](#) describe the constant-product design and the economics of liquidity supply in these markets.

Blockchain logs record each pool call in execution order. We recover a route by following the directed token flows within a transaction. Calls with a common source and destination form a split order; a token received in one call and spent in a later call joins the calls into a sequential route. The same rule separates unrelated conversions bundled in one transaction. For a reconstructed sequence $A \rightarrow B \rightarrow C$, (A, C) is the pair, $A \rightarrow B$ and $B \rightarrow C$ are its legs, and the full sequence is the route. All three objects are directed by token flow. Each leg executes in a liquidity pool,

²Ethereum records a transaction’s ordered logs in its receipt; see the official [Ethereum JSON-RPC documentation](#).

and some pairs are connected only through a vehicle. The observed route endpoints belong to the connected pool component and may differ from the endpoints of a broader user instruction.

A transaction on January 10, 2026 illustrates both the information in the logs and its limit. A public explorer describes a 460,000 PYUSD–USDe swap executed through 0x.³ The user’s instruction begins with PYUSD, but the connected pool sequence begins when 459,929.81 USDC enters Fluid. Fluid returns 460,331.05 USDT, which then enters Uniswap v4 and produces 460,104.28 USDe. The observed pair is therefore USDC $\rightarrow$ USDe; its legs are USDC $\rightarrow$ USDT and USDT $\rightarrow$ USDe, and USDT is the vehicle. The earlier PYUSD $\rightarrow$ USDC conversion occurs outside the connected pool sequence. The role of USDC in that earlier conversion remains unobserved: it may be another vehicle or executor inventory. We restrict reconstruction to the connected pool sequence and leave nearby transfers unassigned. This choice may understate the intermediary role of assets such as USDC when a broader instruction is funded outside the observed pool sequence.

The sequence identifies intermediary use directly. Reconstructing each pool immediately before execution also lets us compose exact-input quotes along the observed route and feasible alternative paths at the same notional. Appendix Equation (6) gives the route composition, Appendix B gives each pool-family quote, and Section 5 applies them to stablecoin and WETH paths.

This intermediary role is distinct from invoicing and funding (Gopinath et al., 2020; Gopinath and Stein, 2021; Mukhin, 2022). It makes the classical volume–cost mechanism observable at transaction level: use expands trading on the two supporting legs, while deeper markets can lower the cost of using the same vehicle again (Krugman, 1980; Kiyotaki and Wright, 1989; Dowd and Greenaway, 1993). Liquidity placement, venue access, and available paths provide the corresponding coordination forces on decentralised exchanges.

2.2 Measuring vehicle dominance

An exact two-leg route has one pool trade on each side of a unique intermediary and therefore one mutually exclusive vehicle choice. A longer route can use several intermediaries. We retain both objects: the native-versus-stable decomposition uses exact two-leg routes, while the wider time series counts every intermediary position and also records whether an asset appears anywhere inside the route.

Routes are reconstructed at the token level before tokens are assigned to economic groups. The native group consolidates ETH with its one-for-one wrapped representation, WETH; liquid-staking tokens remain separate. Dollar-pegged tokens also remain separate instruments because issuer, backing, redemption, risk, and leg-level liquidity differ (Gorton and Zhang, 2023; Catalini et al., 2022; Lyons and Viswanath-Natraj, 2023). The stable group then aggregates those token-level instruments for the family comparison, while issuer results retain USDC, USDT, DAI, and other stablecoins separately.

On each day, the count measure is stablecoin routes divided by all exact two-leg routes whose intermediary is native or stable. The value measure weights the same choices by the mean dollar

³Transaction [0xbda1d07f...9bf67ad9](#).

value at the observed source and destination. It retains only routes for which source, intermediary, and destination values agree within 20%; every intermediary’s incoming and outgoing values must also lie within that band. Appendix Equation (9) gives the exact shares. Levels are percentages, and absolute changes are in percentage pointpercentage points (pppp). Staked-native, imported, and other intermediaries remain in Figure 1 and Appendix Table 25; they stay outside the native-versus-stable denominator.

2.3 The route panel

The route panel covers Curve, Uniswap v1 through v4, Balancer, SushiSwap v2 and v3, and Fluid on Ethereum. It spans 2,798 dates from November 2018–June 2026 and includes constant-product, concentrated-liquidity, stable-swap, weighted-pool, singleton, and integrated-liquidity designs. Indexed blockchain records supply eight exchange series, Dune supplies Fluid, and direct Ethereum queries supply block headers, ordered logs and receipts, token precision, registries, and state checks. Appendix Table 18 reports coverage relative to Ethereum exchange volume.

The source contains 475,071,108 pool swaps and yields 474,388,425 usable directed legs after event direction and token units are standardised. Factory records usually identify pool tokens; for Uniswap v1, the immutable exchange registry supplies the token address and matched ETH flows order the two forced legs. Within every transaction, the directed-flow rule distinguishes single-pool trades, split orders, sequential routes, and unrelated calls. Appendix A gives the reconstruction details.

Transactions with more than one disconnected pool-event group pose a separate measurement limit because the observed calls leave the broader user instruction unknown (Xi and Moallemi, 2026). The principal sample retains transactions with a single connected group. Appendix Table 9 treats each internally valid component as a separate route and finds a slightly larger stablecoin rotation.

Every non-self-returning sequence retained by the connected-transaction rule enters the count measure. Dollar-weighted results add the 20% value-agreement rule, which selects routes with a coherent dollar notional. Appendix F reports the separate round-trip exclusion, and each value exhibit reports its sample coverage.

Transactions and pool-state changes are publicly observable (Schär, 2021). The reconstruction identifies each executed route and its calling contract; the economic unit remains the executed route because an executor address can serve several frontends or quoting systems (Xi and Moallemi, 2026).

Full-day reconciliation against Ethereum logs covers 206 venue-days across 74 dates. The Uniswap v2, SushiSwap v2, and Uniswap v3 comparisons span 13.9 million indexed swaps. Precision is 100.0000%, 100.0000%, and 99.9998%, respectively; recall is 99.9995%, 100.0000%, and 99.9335%. Reconstructing the corrected days changes stablecoin share by -0.008 pp by route count and -0.016 pp by routed value. Every component of the sampled 2024–2026 decomposition moves by at most 0.001 pp. Appendix Table 10 reports this reconciliation, Appendix A gives the remaining construction checks, and Appendix B validates the pretrade pricing models.

Table 1: The route panel

	Route panel
Dates in the full route panel	2,798
Missing source-days	0
Raw swap rows	475,071,108
Usable directed route legs	474,388,425
Routes per day, median across eligible days	153,586
Multi-leg routes per day, median across eligible days	30,064
Multi-leg share of routes, median across eligible days	19.4%
Round trips, median share of multi-leg routes by count	12.8%
Round trips, median share of multi-leg routes by value	21.0%

Note: The full 9-deployment panel contains UTC dates from November 2018–June 2026. Uniswap v1 enters these totals through the exact exchange-to-token registry; its protocol-architecture comparison is venue-specific. Daily route statistics and round-trip shares use all 2,449 eligible dates from 15 May 2019 through 30 June 2026. A multi-leg route traverses more than one pool; a round trip ends in the token with which it began. Round-trip shares are formed among multi-leg routes.

3 How stablecoins gain the vehicle role

3.1 The aggregate rotation

Stablecoins carry a much larger share of vehicle activity in 2026 than in 2024. The comparison uses the same January–June dates in both years because the 2026 panel ends on June 30. Giving each matched day equal weight, stablecoins’ share among native and stable intermediaries rises from 16.9% to 42.1% by route count and from 32.7% to 76.5% by routed value. Appendix Table 21 reports the paired-day estimates and their sampling uncertainty. The decomposition below pools route activity so its components allocate the aggregate count or value exactly.

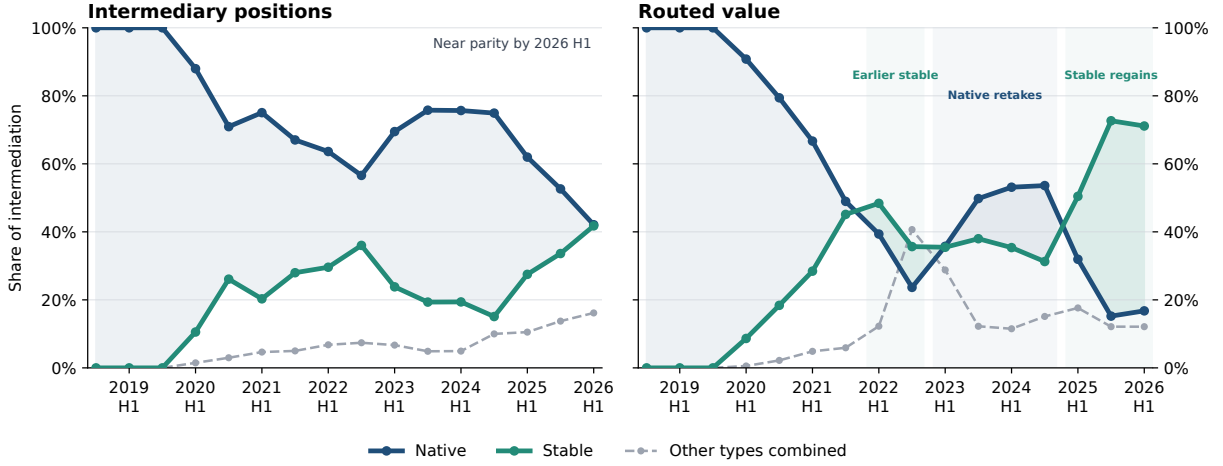
The half-year series in Figure 1 reverses direction. Native assets regain share in 2023 and 2024; stablecoins recover the routed-value lead in 2025. The issuer results below distinguish USDT and USDC within that family.

3.2 Pair formation and persistence

Aggregate dominance can change inside continuing pairs or through the distribution of activity across pairs. Let p denote a pair and $y \in \{2024, 2026\}$. A continuing pair carries native-or-stable routes in both matched periods; a year-specific pair carries them in only one. First split total activity between these two sets. Within continuing pairs, stablecoin share can change because a given pair changes its vehicle mix or because trading shifts toward pairs with a different vehicle mix. The remaining change comes from the aggregate weight on continuing pairs and the stablecoin share of pairs observed in only one period.

The exact midpoint decomposition holds each companion quantity at its two-period average while the other changes. Its four terms add to the pooled rotation: vehicle switching within continuing pairs, activity shifting across those pairs, the aggregate weight on continuing pairs, and

Figure 1: Intermediary activity rotates between native assets and stablecoins



Note: The sample contains complete non-self-returning routes. Each route contributes once for every intermediary it uses, so routes with more than two legs can contribute several observations. Shares are formed across all intermediary types; Appendix Table 21 uses the narrower native-or-stable denominator. Every point covers six calendar months; the final point is 2026 H1. Routed value requires source, intermediary, and destination dollar amounts to agree within 20%.

the stablecoin share of pairs present in only one period. We also compare the same pair, month-day, and realised single- or cross-exchange route class with pair-by-calendar-cell effects. Appendix G gives the algebra and matched specification.

Table 2 applies both comparisons to route counts and supported value.

Pooling route counts gives an increase from 16.9% to 42.4%, or +25.5 pp, close to the equal-day change in Appendix Table 21. The within-pair component is -0.1 pp by route count and 0.0 pp by routed value. These net estimates combine +1.3 pp from pairs moving toward stablecoins and -1.4 pp from pairs moving toward the native asset. Activity shifting across continuing pairs contributes +8.4 pp by count and +26.2 pp by value. The changing aggregate weight on continuing pairs contributes -0.5 pp and -2.5 pp (Table 2, panels A and B).

The period-specific contribution is also a net. Pairs first observed after 2024 H1 contribute +20.1 pp by route count and +21.9 pp by routed value. Reactivation, vehicle-role turnover, and exit together contribute -2.3 pp and -2.8 pp, leaving the period-specific terms at +17.8 pp and +19.2 pp. The underlying rows add exactly before rounding. First-observed pair entry therefore supplies 78.9% of the total route-count rotation and 51.2% of the dollar-weighted rotation. The matched specification reaches the same conclusion on the more restrictive sample: stablecoin share changes by +0.2 pp by count and -1.4 pp by value (Table 2, panel C). A large aggregate rotation therefore coexists with -0.1 pp net switching inside established cells defined by pair, calendar date, and route class.

The pattern extends through time and across endpoint sets. In every adjacent January–June comparison from 2019 through 2026, the within-pair term lies between -0.4 pp and +1.0 pp even as the aggregate moves from -6.4 pp in 2022–2023 to +17.4 pp in 2025–2026 (Appendix Table 23,

Table 2: Decomposition of the stablecoin rotation

Component or estimate	Estimate [pp]	Obs.
<i>Panel A. Route-count share: continuing and year-specific pairs</i>		
Net stablecoin-share change within continuing pairs	−0.1	
Pairs moving toward stablecoins (1,569)	+1.3	
Pairs moving toward native assets (1,487)	−1.4	
Vehicle activity shifting across continuing pairs	+8.4	
Weight of continuing versus year-specific pairs	−0.5	
Net contribution of period-specific vehicle activity	+17.79	
Pairs first observed after 2024 H1	+20.09	
Pairs reactivated after absence in 2024 H1	+0.20	
Vehicle-role turnover in continuing pairs	−0.77	
Pairs exiting before 2026 H1	−1.73	
Total route-count change	+25.5	
<i>Panel B. Dollar-weighted share: continuing and year-specific pairs</i>		
Net stablecoin-share change within continuing pairs	0.0	
Pairs moving toward stablecoins (1,505)	+2.3	
Pairs moving toward native assets (1,445)	−2.4	
Vehicle activity shifting across continuing pairs	+26.2	
Weight of continuing versus year-specific pairs	−2.5	
Net contribution of period-specific vehicle activity	+19.16	
Pairs first observed after 2024 H1	+21.94	
Pairs reactivated after absence in 2024 H1	+0.04	
Vehicle-role turnover in continuing pairs	−1.72	
Pairs exiting before 2026 H1	−1.10	
Total change in dollar-weighted share	+42.8	
<i>Panel C. Matched pair estimates</i>		
All two-leg routes, count share	+0.23 (0.77)	188,344
20% agreement sample, count share	+0.32 (0.75)	182,734
20% agreement sample, dollar-weighted share	−1.35 (2.19)	182,734

Note: Panels A and B compare pooled activity over matched January–June dates in 2024 and 2026 and report the exact midpoint decomposition in Appendix Equation (10). Their first indented rows split the within-pair component into positive and negative contributions. Their lifecycle rows split the net contribution of period-specific vehicle activity using each pair’s first and last observation in the full panel: first-observed entry, reactivation after absence in 2024 H1, vehicle-role turnover among pairs active in both periods, and exit before 2026 H1. These lifecycle rows add to the net term using unrounded values. Panel B weights routes by dollar value and applies the 20% agreement rule. Panel C estimates Appendix Equation (11); standard errors use two-way clustering by pair and date. Appendix Table 22 reports an alternative factorisation of the route-count change.

panel A). Removing every pair with WETH or a stablecoin at either endpoint still gives an increase from 1.1% to 9.0% by count and from 0.9% to 39.1% by supported value (panel B). Appendix Table 24 then allocates the headline change across endpoint directions and shows that USDT supplies +13.7 pp of the routed-value increase between two stablecoin endpoints. Pair composition therefore explains stablecoin gains and reversals even when endpoint identity leaves both broad vehicle families available.

Across matched route cells, adjacent periods, and endpoint sets, activity moving among pairs accounts for most of the aggregate change. Appendix G gives the decomposition algebra and reports the adjacent-period, endpoint, issuer, and network results.

4 How vehicle use becomes established

The pair decomposition locates the aggregate rotation in pairs that become active and attract trading. We first ask whether the vehicle used when a pair enters persists. We then study how two-leg capital makes a competing vehicle available and how exact prices allocate trading once stablecoin and WETH paths coexist.

4.1 Vehicle persistence after pair entry

The vehicle used when a pair enters strongly predicts its subsequent use. For pair p , let E_p be the stablecoin share on its entry day. For each of the disjoint windows $h \in \{1-30, 31-120\}$, let $R_{p,h}$ indicate that the pair carries another native- or stablecoin-intermediated route, and let $S_{p,h}$ be the stablecoin share when $R_{p,h} = 1$. We estimate

$$\begin{aligned} S_{p,h} &= \alpha_h + \beta_h E_p + X_p' \delta_h + \varepsilon_{p,h}, & R_{p,h} &= 1, \\ R_{p,h} &= a_h + b_h E_p + X_p' c_h + u_{p,h}, & \text{all entrant pairs.} \end{aligned} \tag{1}$$

The second line is a linear probability model. The controls X_p contain the cohort year, a stablecoin-endpoint indicator, log entry-day route count, direct-route share, and complex-route share. The sample uses pairs entering by March 2 in 2024 or 2026, giving every pair 120 complete calendar days of follow-up, and keeps pairs whose endpoints lie outside the native group.

We separate later vehicle use from pair survival. Of the 157,262 entrants, 19.4% trade again during days 1–30 and 12.3% do so during days 31–120 (Table 3, panel B). Among pairs that trade again, a 10 pp higher entry-day stablecoin share predicts +8.92 pp higher stablecoin share during days 1–30 and +8.40 pp during days 31–120 (panel A, columns 2 and 5). The route-activity-weighted estimates are +8.55 pp and +9.00 pp, respectively (columns 3 and 6). Every follow-up outcome begins after the entry day, so these estimates capture continued vehicle use across subsequent routes.

Entry-day stablecoin share also predicts a modestly higher probability of subsequent trading: a 10 pp increase corresponds to +0.30 pp during days 1–30 and +0.98 pp during days 31–120

Table 3: Vehicle persistence after pair entry

<i>Panel A. Stablecoin route share among pairs that trade again</i>						
	Days 1–30			Days 31–120		
	(1)	(2)	(3)	(4)	(5)	(6)
Entry stablecoin share [effect pp per 10 pp]	+9.00*** (0.14)	+8.92*** (0.15)	+8.55*** (0.72)	+8.59*** (0.18)	+8.40*** (0.18)	+9.00*** (0.50)
Controls	No	Yes	Yes	No	Yes	Yes
Route-activity weights	No	No	Yes	No	No	Yes
Observations (pairs)	30,547	30,547	30,547	19,405	19,405	19,405
Entry-date clusters	123	123	123	123	123	123
Mean stablecoin share [pp]	2.70	2.70	4.78	3.83	3.83	7.18
R^2	0.842	0.843	0.800	0.787	0.791	0.830

<i>Panel B. Subsequent-trading incidence among all entrants</i>		
	Days 1–30	Days 31–120
Entry stablecoin share [effect pp per 10 pp]	+0.30*** (0.08)	+0.98*** (0.08)
Controls	Yes	Yes
Observations (pairs)	157,262	157,262
Entry-date clusters	123	123
Mean retrading rate [pp]	19.42	12.34
R^2	0.068	0.033

Note: Panel A estimates the first line of Equation (1) among pairs that carry another native- or stablecoin-intermediated route in the stated window. Panel B estimates the second line on all entrant pairs. The entry day is excluded from both follow-up windows. Controls are the cohort year, a stablecoin-endpoint indicator, log entry-day route count, direct-route share, and complex-route share. Columns 3 and 6 weight pairs by route activity in the respective follow-up window; all other columns give each pair equal weight. Coefficients report the outcome change in percentage points (pp) associated with a 10 pp higher entry-day stablecoin share. Standard errors in parentheses use CR1 clustering by entry date. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

(Table 3, panel B). Vehicle identity therefore persists strongly among pairs active in each window, alongside a smaller survival margin. We interpret both as descriptive equilibrium relationships: relative prices, depth, and venue access can shape the vehicle at entry and remain relevant for later trading.

The wider route panel places this persistence result in its network and protocol setting.

Figure 1 includes every complete route, with one observation for each intermediary position. Stablecoins gain across both short and long routes and within single- and cross-exchange routes. Their all-route participation rises from 17.6% in 2024 to 46.0% in 2026 H1, while WETH remains the highest-betweenness node in every annual trading graph (Appendix Table 25). Connectivity therefore moves less than realised intermediary use; local depth and execution cost determine how useful a connection is for a particular route.

Protocol history gives a benchmark for inherited vehicle use. Uniswap v1 required each token-to-token trade to cross ETH; v2 allowed any two ERC-20 tokens to form a pool.⁴ We reconstruct 217,003 connected v1 token-to-token routes. In 2026, after six years of open pair formation, 95.5% of single-leg v2 trades still execute in WETH pools and 97.9% of newly traded token combinations include WETH (Appendix Table 11, panels A and B). Venue migration changed the available token set, so this chronology serves as descriptive context. The bridge and price estimates below compare pairs in which stablecoin and WETH routes coexist.

⁴See Adams, Zinsmeister, and Robinson, *Uniswap v2 Core*, pp. 1–2.

4.2 Full-range constant-product capital and route allocation

The classical vehicle-currency mechanism links trading volume to lower transaction costs (Krugman, 1980). Here an indirect route needs two legs: one connecting the source to the vehicle and another connecting the vehicle to the destination. We measure deposited capital as validated full-range reserve value in Uniswap v2 and SushiSwap v2 pools. The weaker leg determines this two-leg capital measure. It is an equilibrium stock on those constant-product venues; size-specific pretrade quotes provide the closer measure of executable liquidity across the broader venue set. Concentrated-liquidity providers can also change executable depth by moving capital along the price curve (Adams et al., 2021; Lehar and Parlour, 2025).

We date bridge formation using only information available before route choice. The event is the first date on which DAI, USDC, or USDT has at least \$10,000 of prior-calendar full-range capital in Uniswap v2 and SushiSwap v2 pools on each of the two legs required by the pair. Every pair routed through WETH earlier, and the event predates any observed stablecoin route in the pair. This definition yields 1,618 bridge events. Table 4, panel A, shows that 38.3% carry a supported stablecoin route within 30 days and 47.0% do so within 120 days. Among the first-month adopters, 62.4% use a supported stablecoin again during days 30–119, although stablecoins account for only 8.2% of their route activity in that later window. The two-leg capital threshold therefore precedes adoption, while initial use often leaves WETH as the principal vehicle.

The capital comparison follows each bridge after support appears. Let $K_{av,t-1}$ denote prior-calendar full-range capital in Uniswap v2 and SushiSwap v2 pools on the leg joining endpoint a to vehicle v . The stablecoin set is $\mathcal{S} = \{\text{DAI}, \text{USDC}, \text{USDT}\}$. For pair (i, j) , define

$$\begin{aligned} D_{et}^S &= \max_{v \in \mathcal{S}} \min\{K_{iv,t-1}, K_{vj,t-1}\}, & D_{et}^W &= \min\{K_{iW,t-1}, K_{jW,t-1}\}, \\ Q_{et} &= \frac{D_{et}^S}{D_{et}^S + D_{et}^W}, & S_{et}^R &= \beta Q_{et} + \alpha_e + \lambda_{m(t)} + \eta_{a(t)} + \varepsilon_{et}. \end{aligned} \quad (2)$$

Here W denotes WETH, S_{et}^R is stablecoin route share, $m(t)$ is calendar month, and $a(t)$ groups days since support into seven-day intervals. Event effects compare changes in relative full-range constant-product capital within the same pair and route scope.

Panel B compares route allocation after that prior-information event with the preceding 30 days. Stablecoin route share rises by +5.60 pp during days 0–29 and by +5.49 pp during days 30–119. Panel C uses the continuous relative-capital share within the same event. A 10 pp increase in the stablecoin share of prior-day weak-leg full-range capital is associated with +6.90 pp more stablecoin route activity during days 0–29 and +8.35 pp during days 30–119. The later coefficient is larger even though adoption remains incomplete. The threshold marks full-range support on both legs; relative capital continues to divide trading after the bridge appears.

The threshold event and subsequent relative-capital estimates connect pair formation to full-range liquidity on Uniswap v2 and SushiSwap v2 using information available at the event date. Appendix Table 27 and Figure 3 follow the alternative persistent-support and first-use timing defi-

Table 4: Prior full-range constant-product capital, stablecoin adoption, and route allocation

Quantity	Estimate	Events
<i>Panel A. Route use after the lagged-capital threshold</i>		
First supported stablecoin route within 30 days	38.3%	1,618
First supported stablecoin route within 120 days	47.0%	1,618
Stablecoin used during days 30–119, among first-month adopters	62.4%	620
Stablecoin route share during days 30–119, same pairs	8.2%	620
<i>Panel B. Change from the prior 30 calendar days</i>		
Stablecoin route share, days 0–29 [pp]	+5.60*** (1.01)	1,618
Stablecoin route share, days 30–119 [pp]	+5.49*** (1.09)	1,618
<i>Panel C. Prior-day weak-leg depth and route allocation</i>		
10 pp higher stablecoin share of relative depth, days 0–29 [pp]	+6.90*** (0.70)	1,495
10 pp higher stablecoin share of relative depth, days 30–119 [pp]	+8.35*** (0.43)	1,105

Note: The event is the first date on which DAI, USDC, or USDT has at least \$10,000 of prior-calendar full-range reserve value in Uniswap v2 and SushiSwap v2 pools on both route legs. Every pair used WETH earlier, and the event predates any observed stablecoin route in the pair. Panel A follows the stablecoin or stablecoins that cross the threshold on the event date. Panel B reports activity-weighted changes from the prior 30 calendar days with pair- and event-date-clustered standard errors. Panel C estimates Equation (2) with bridge-event effects, calendar-month effects, seven-day event-age controls, and pair- and date-clustered standard errors. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. This capital measure is an equilibrium stock on the two constant-product venues; the exact-price comparisons use the broader supported venue set.

nitions. They show that most pre-use capital growth occurs in pools already active one week earlier (Appendix Table 28). This pattern is suggestive of providers deepening a bridge in anticipation of route demand, traders waiting for usable depth, or both forces operating together. Liquidity supply is voluntary and can respond to expected trading conditions (Li et al., 2021). The post-entry persistence estimates remain similar among pairs with more entry-day trades (Appendix Table 26). When both vehicles become executable, current exact prices and lagged full-range capital jointly predict the selected vehicle.

5 Exact prices, full-range capital, and vehicle retention

Executable price differences can persist when trading is segmented across cryptocurrency markets (Makarov and Schoar, 2020). We therefore compare stablecoin and WETH routes at the same input and pretrade state. For each observed two-leg route on the fifteenth day of every month from June 2020 through June 2026, we quote the best WETH route and the best stablecoin route through DAI, USDC, or USDT across Uniswap v2, SushiSwap v2, and Uniswap v3. These venues carry 93.4% of linear routes on the sampled dates, and 89.2% of mapped observed routes reproduce realised output within one basis point. Both vehicle families must be feasible; every observed and hypothetical leg must remain below 5% own-price impact; leg values must agree within 20%; and input value must be at least \$100. Appendix B gives the pricing equations, Appendix D calibrates the common support rule, and Appendix Table 16 reports the broader price frontier.

5.1 The first exact contest after pair entry

The entry-day estimates show persistence across later realised routes, including pairs for which only one vehicle family may be feasible. We now locate the first sampled monthly date after pair entry on which the same observed trade can use either WETH or a stablecoin path under the exact common-support rules. Pair entry requires at least \$5,000 of coherent route value on its first active day. Among 118,447 such entrants, 580, or 0.5%, reach a sampled exact contest. The median lag is 115.5 days, and 53.3% reach it within 120 days (Table 5, panel A). The estimates describe this strict monthly opportunity set.

For route r on pair p 's first contestable date t , we estimate

$$C_{rpt} = \alpha + \beta A_{rpt}^O + \gamma A_{pt-1}^K + \beta_q \log q_r + \delta_t + \eta_i + \xi_j + \phi_s + \varepsilon_{rpt}. \quad (3)$$

In column 1 of panel B, C_{rpt} indicates stablecoin use and A_{rpt}^O is the stablecoin's exact-output advantage. In columns 2 and 3, C_{rpt} indicates retention of the entry family, while A_{rpt}^O and A_{pt-1}^K are oriented toward that family. The capital term is its share of prior-calendar full-range constant-product weak-leg capital. Date effects δ_t , source effects η_i , destination effects ξ_j , and observed-route-scope effects ϕ_s compare routes inside the same dated opportunity structure.

The entry family carries 83.1% of routes at the first sampled exact contest; equal weighting

across pairs gives 84.4% (Table 5, panel A). Across all first-contest routes, the coefficient on a 100 bp stablecoin exact-output advantage is +6.39 pp (panel B, column 1). On the common capital sample, the coefficient on a 100 bp exact-output advantage for the entry family is +10.31 pp in column 2 and +10.20 pp after prior capital enters in column 3. The prior-capital coefficient is +1.85 pp per 10 pp, with a standard error of 4.58 pp (column 3). At the first contest, executable price is the precise predictor; the separate prior-capital coefficient is imprecise. In established pairs and at price-rank crossings, prior capital predicts mature retention and the durability of a new price lead.

Table 5: Vehicle choice at the first sampled exact contest

Panel A. From original pair entry to the first sampled exact contest			
Quantity	Primary value		Additional detail
Original pair-entry cohort	118,447 pairs	2,090 entry dates; \$5,000 minimum	
First sampled exact contest	580 pairs	995 routes	
Entry pairs reaching a sampled contest	0.5%		
Entry-to-contest lag [days]	115.5 median	21–318.5 interquartile range	
Contest reached within 120 days	53.3%		
Entry family retained, clear comparisons	83.1% of routes	84.4% of pairs	
Positive prior-day V2 capital for both families	220 routes	115 pairs	
Panel B. Route choice at the first sampled exact contest			
	(1)	(2)	(3)
Outcome; effects [pp]	Stablecoin chosen	Entry family retained	Entry family retained
Stablecoin current exact-output advantage [per 100 bp]	+6.39*** (0.75)		
Entry-family current exact-output advantage [per 100 bp]		+10.31** (3.93)	+10.20** (4.16)
Entry-family prior-day V2 weak-leg capital share [per 10 pp]			+1.85 (4.58)
Log route input value [per log point]	−2.51 (1.61)	−4.01 (4.90)	−3.91 (5.01)
Dependent mean [%]	22.1	84.9	84.9
Within R^2	0.162	0.152	0.152
Observations (routes)	995	219	219
Ordered endpoint pairs	580	114	114
Dates	64	27	27
Prior-day V2 capital positive for both families	No	Yes	Yes
Date, source, destination, and route-scope fixed effects	Yes	Yes	Yes
Pair- and date-clustered s.e.	Yes	Yes	Yes

Note: Original pair entry is the first observed activity with at least \$5,000 of coherent route value for a pair. The first sampled exact contest is the first monthly exact-state date after entry on which both a WETH path and a path through DAI, USDC, or USDT are feasible for an observed route under the common quote rules. Panel A describes the resulting lifecycle and retains only pairs with a clear native or stablecoin entry family for the retention rows. Exact states are sampled monthly and the common quote rules are deliberately strict. The 0.5% coverage rate refers to this sampled set; lifetime feasibility across all entrants remains outside measurement. Panel B estimates Equation (3). Exact output comes from the pre-transaction state and its advantage is capped at 1,000 bp in absolute value. Capital is prior-calendar full-range Uniswap v2 and SushiSwap v2 capital on the weaker route leg; its regressor is the entry family’s share of the two families’ capital, centred at 50%. Every model includes log input value and date, source, destination, and observed-route-scope effects. Standard errors are clustered by pair and date. Coefficients and standard errors are in pp. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

5.2 Prices and capital across established pairs

We call a family the incumbent when it alone carries the pair on the first observed vehicle day and begin measuring retention 30 days later. For route r , pair p , and date t , the price-and-capital specification is

$$R_{rpt} = \beta_O A_{rpt}^O + \beta_K A_{pt-1}^K + \beta_q \log q_r + \alpha_p + \delta_t + \varepsilon_{rpt}. \quad (4)$$

Here R_{rpt} equals one when the route retains the incumbent family. The exact-output advantage is $A_{rpt}^O = [10^4(O_{rpt}^{\text{inc}} - O_{rpt}^{\text{chall}})/O_{rpt}^{\text{chall}}]_{-1000}^{1000}/100$, so one unit represents 100 basis points. For family v , weak-leg capital is $K_{pt-1}^v = \min\{K_{i,v,t-1}, K_{v,o,t-1}\}$, where each leg aggregates prior-calendar deposited capital across full-range Uniswap v2 and SushiSwap v2 pools. Its relative advantage is $A_{pt-1}^K = \{K_{pt-1}^{\text{inc}}/(K_{pt-1}^{\text{inc}} + K_{pt-1}^{\text{chall}}) - 1/2\}/0.10$, so one unit represents a 10-percentage-point shift from an equal capital split. Pair fixed effects α_p absorb persistent pair-level routing preferences, and date fixed effects δ_t absorb market-wide conditions.

Table 6 reports the route-choice estimates. Column 1 relates contemporaneous price leadership to route choice; the coefficient on stablecoin price leadership is +57.59 pp. Among mature pairs with an exclusive incumbent, the incumbent carries 93.3% of routes when it leads on exact output and 27.2% when the challenger leads. In column 2, the within-pair, within-date coefficient on challenger leadership is -58.08 pp. Columns 3 and 4 use the same 17,778 routes, each with positive prior-day full-range constant-product capital for both vehicle families. The coefficient on a 100-basis-point incumbent output advantage is +10.56 pp when entered alone in column 3 and +10.13 pp after capital enters in column 4. A 10-percentage-point incumbent capital-share advantage is associated with +2.77 pp higher retention in column 4. The exact-output coefficient changes little across the two columns, while lagged two-leg capital adds explanatory power beyond the current quote.

Table 6: Exact prices, full-range weak-leg capital, and vehicle retention

Outcome; estimates [pp]	(1) Stablecoin chosen	(2) Incumbent retained	(3) Incumbent retained	(4) Incumbent retained
Stablecoin route has higher exact output	+57.59*** (2.84)			
Challenger route has higher exact output		-58.08*** (2.82)		
Challenger output lead $\times$ stable incumbent		+4.47 (7.82)		
Incumbent exact-output advantage [100 bp]			+10.56*** (0.83)	+10.13*** (0.77)
Incumbent lagged full-range capital-share advantage [10 pp]				+2.77*** (0.61)
Log input value [USD]	+0.32 (0.43)	-0.66 (0.46)	-1.56* (0.87)	-1.32* (0.78)
Dependent mean [%]	26.5	73.7	69.7	69.7
Within R^2	0.372	0.382	0.145	0.153
Observations	49,212	34,439	17,778	17,778
Pair clusters	2,596	1,981	678	678
Date clusters	73	73	73	73
Pair fixed effects	Yes	Yes	Yes	Yes
Date fixed effects	Yes	Yes	Yes	Yes
Two-way clustered s.e.	Pair, date	Pair, date	Pair, date	Pair, date
Exclusive entry, age ≥ 30 days	No	Yes	Yes	Yes
Prior-day full-range capital positive, both vehicles	No	No	Yes	Yes
Minimum absolute output difference [bp]	1	1	1	1
Continuous output gap cap [bp]			1,000	1,000

Note: The unit of observation is a realised two-leg route on the fifteenth day of each month from June 2020 through June 2026. For the same ordered endpoint pair, input amount, and pre-transaction state, we quote the best WETH route and the best stablecoin route through DAI, USDC, or USDT across Uniswap v2, SushiSwap v2, and Uniswap v3. Input value is at least \$100, leg values agree within 20%, both vehicle families are feasible, and every observed and quoted leg remains below 5% own-price impact. Ties within one basis point are omitted, and all prices are gross of gas. Column 1 is a linear probability regression of stablecoin use on an indicator that the stablecoin route returns more exact output. Columns 2 through 4 study retention after a pair first uses one vehicle family exclusively and that first use is at least 30 days old. Column 2 includes an indicator that the challenger returns more exact output and its interaction with a stablecoin incumbent. Columns 3 and 4 use the same observations with positive prior-day full-range constant-product weak-leg capital for both families; column 3 includes the continuous output advantage, and column 4 adds the capital-share advantage in Equation (4). Weak-leg capital is the smaller deposited-capital stock across the two required legs, aggregated across Uniswap v2 and SushiSwap v2; the capital-share regressor measures the incumbent's deviation from one-half. The exact-output advantage is measured per 100 basis points and capped at plus or minus 1,000 basis points; the capital-share advantage is measured per 10 percentage points; input value enters in logs. Every column includes ordered endpoint-pair and date fixed effects. Standard errors in parentheses are clustered by pair and date. Coefficients and standard errors are in percentage points. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

5.3 When the price lead changes hands

The retention regressions compare routes across many price configurations. We next follow the same pair when its exact-output ranking reverses. A price-rank crossing occurs when a challenger moves from at least one basis point behind in month $t - 1$ to at least one basis point ahead in month t . The ranking uses the pair-month median exact-output gap across common-support quotes. Event dating uses only months $t - 1$ and t . The principal comparison requires at least two common-support routes and at least \$1,000 of observed input in each crossing month. It contains 400 crossings across 180 pairs, split equally between stablecoin and WETH challengers.

Let $R_{e\tau}$ be the incumbent family’s route share in event e and relative month τ , and let $Q_{e,-1}$ be the challenger’s share of prior-calendar full-range constant-product weak-leg capital on the eve of the crossing. We estimate

$$\begin{aligned} R_{e0} - R_{e,-1} &= \alpha + \gamma Q_{e,-1} + X_e' \delta + \lambda_{m(e)} + \varepsilon_e, \\ L_{e,1} &= a + \theta Q_{e,-1} + X_e' c + \lambda_{m(e)} + u_e, \end{aligned} \tag{5}$$

where $L_{e,1}$ equals one when the challenger remains the exact-output leader one month later. The controls X_e contain the challenger family, the exact-output gaps in months $t - 1$ and t , prior incumbent route share, and log route activity across the two crossing months. Calendar-month effects $\lambda_{m(e)}$ absorb common market conditions.

Among the 147 crossings observed from month -3 through month $+3$, the incumbent carries 66.9% of routes immediately before the crossing and 38.1% in the crossing month (Table 7, panel A). One month later, its share is 49.2%. Incumbent share falls 29.0 pp more at the crossing than during the same event’s earlier move from month -3 to month -2 (panel B, column 4). Route allocation therefore moves sharply when the sampled pair-month exact-output ranking changes.

The next month separates a temporary quote from a durable challenge. The challenger remains ahead in 50.6% of observed events, and the incumbent retakes the lead in 46.5%. Incumbent route share is 23.0% in the first group and 73.4% in the second. A 10 pp larger challenger share of prior weak-leg capital predicts 3.70 pp greater probability that the challenger remains ahead one month later (Table 7, panel B, column 2). Its immediate route-share coefficient in column 1 is small. Exact output tracks immediate allocation; prior two-leg capital predicts whether the new price lead persists.

Table 7: Price-rank crossings, route reallocation, and weak-leg capital

Panel A. Incumbent route share around an exact-price rank crossing [pp]				
Event month	All crossings	Stable challenger	Native challenger	
-3	54.7 (2.0)	73.7 (3.0)	35.4 (3.6)	
-2	54.8 (2.4)	73.6 (3.5)	35.7 (4.6)	
-1	66.9 (1.8)	83.1 (2.7)	50.5 (3.8)	
0 (crossing)	38.1 (1.9)	53.6 (3.8)	22.5 (2.7)	
+1	49.2 (2.5)	69.2 (4.3)	29.0 (4.4)	
+2	55.4 (2.3)	77.9 (4.2)	32.5 (4.2)	
+3	50.4 (2.5)	73.4 (3.6)	27.0 (3.9)	
Panel B. Immediate response and next-month price-rank persistence				
	Incumbent-share change	Challenger leads next month		Actual vs. placebo
	Activity floor (1)	Activity floor (2)	All crossings (3)	Activity floor (4)
Challenger weak-leg capital share, $Q_{e,-1}$ [10 pp]	+0.28 (0.66)	+3.70** (1.52)	+3.85*** (0.95)	
Actual crossing [vs. months -3 to -2]				-28.98*** (4.71)
Actual crossing $\times$ challenger capital [10 pp]				+2.75 (1.87)
Crossing controls	Yes	Yes	Yes	Yes
Crossing-month fixed effects	Yes	Yes	Yes	No
Crossing-event fixed effects	No	No	No	Yes
Observations	361	286	583	440
Crossing events	361	286	583	220
Ordered endpoint pairs	163	122	229	83

Note: A price-rank crossing occurs when the pair-month median exact-output gap moves from at least one basis point in favour of the incumbent in month $t - 1$ to at least one basis point in favour of the challenger in month t . The principal activity floor requires at least two common-support routes and \$1,000 of observed input in both crossing months. Panel A gives equal-event means for the 147 crossings observed throughout months -3 to $+3$; parentheses contain ordered endpoint-pair-clustered standard errors. In panel B, column 1 estimates the first line of Equation (5); columns 2 and 3 estimate its second line on the principal and unrestricted crossing samples. These columns control for challenger family, crossing- and prior-month exact-output gaps, prior incumbent route share, and log crossing-month route activity and include crossing-calendar-month effects. Standard errors are clustered by pair and calendar month. Column 4 stacks each event's month -1 to month 0 change with its month -3 to month -2 change and includes event effects; standard errors are clustered by pair. Weak-leg capital is the smaller prior-calendar full-range capital stock across the two required legs, aggregated over Uniswap v2 and SushiSwap v2. Coefficients and standard errors are in pp. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. Deposited capital is an equilibrium pool stock, and pair-month prices are medians across observed trade sizes.

The same comparison gives the output consequence of vehicle choice. Panel A of Table 8 shows that 12.9% of routes use the family that returns at least one basis point less. Conditional on that choice, the median shortfall is 27.2 bp and the 90th percentile is 171.5 bp; weighting every contestable route by input value gives 7.4 bp. Among routes that retain a mature exclusive incumbent, 10.7% leave more than one basis point of gross output, with a 17.1 bp conditional median and 3.3 bp across all retained routes (panel B). Panel C reports value-weighted shortfalls of 16.8 bp for pairs aged 0–89 days, 19.0 bp for pairs aged 90–364 days, and 2.1 bp after one year.

Panel D matches the exact-route comparison to transaction receipts and predicts gas separately for the chosen and rival paths. On the common receipt-matched sample, the input-value-weighted shortfall rises from 7.46 bp gross to 7.65 bp net of gas. For inputs below \$1,000, it rises from 16.02 to 22.01 bp; above \$100,000, the estimates are 3.78 and 3.77 bp. Gas therefore sharpens the consequence for small trades while leaving the aggregate magnitude nearly unchanged. The price cost is concentrated earlier in a pair’s life and at smaller trade sizes.

Appendix Table 16 widens the same-state opportunity set beyond the two named vehicle families. Panel A first opens additional venues for the observed vehicle and then opens every named public vehicle plus the direct route. The final expansion adds only 2.0 pp to the share of routes with a gross-output improvement, while the median improvement across all better routes is 21.9 bp. Panel B shows a route-weighted stablecoin-share change of 1.2 pp. The broader comparison changes the venue classification more often than the intermediary asset.

Receipt gas adds a fixed toll to the extra leg and can affect both trading costs and liquidity concentration (Caparros et al., 2024). Appendix E reports the realised toll over time and the held-out validation behind Panel D’s path-specific predictions.

The benchmark represents public on-chain liquidity in the declared exchange set. Private order flow, financing costs, losses from reverted transactions, and pools outside that set can alter a router’s choice. Across sampled Curve pool-days, pools outside the StableSwap model carry 34.2% of assessed volume, including 65.2% of native-leg volume and 21.1% of stablecoin-leg volume (Appendix Table 19, panel A). Hook-bearing or dynamic-fee Uniswap v4 pools, SushiSwap v3, and several Balancer families also use pricing states outside the benchmark. Appendix E reports the remaining coverage limits.

Table 8: Gross and gas-adjusted output shortfalls across feasible vehicle families

Measure	Estimate	Routes			
<i>Panel A. All contestable routes</i>					
Observed family has an exact-output shortfall above 1 bp [%]	12.9%	52,477			
Conditional median shortfall [bp]	27.2	6,786			
Conditional 90th-percentile shortfall [bp]	171.5	6,786			
Input-value-weighted mean shortfall [bp]	7.4	52,477			
<i>Panel B. Routes retaining a mature exclusive incumbent</i>					
Observed family has an exact-output shortfall above 1 bp [%]	10.7%	27,215			
Conditional median shortfall [bp]	17.1	2,901			
Input-value-weighted mean shortfall [bp]	3.3	27,215			
<i>Panel C. Input-value-weighted mean shortfall by pair age</i>					
Pairs aged 0–89 days [bp]	16.8	13,159			
Pairs aged 90–364 days [bp]	19.0	13,166			
Pairs aged at least 365 days [bp]	2.1	26,152			
<i>Panel D. Receipt-matched routes, gross and gas-adjusted output</i>					
	Routes	Lower-output route [%]	Weighted shortfall [bp]		
Input value		Gross	Net of gas	Gross	Net of gas
All routes	52,207	13.0	13.7	7.46	7.65
\$100–999	21,371	13.2	14.7	16.02	22.01
\$1,000–9,999	22,919	12.8	13.0	9.20	9.51
\$10,000–99,999	7,317	13.1	13.1	9.20	9.21
\$100,000 and above	600	9.7	9.7	3.78	3.77

Note: The unit is a realised two-leg route in the common-support sample underlying Table 6. For the same ordered endpoint pair, input amount, and pre-transaction state, we compare the best route through WETH with the best route through DAI, USDC, or USDT across Uniswap v2, SushiSwap v2, and Uniswap v3. Every observed and quoted leg remains below 5% own-price impact, and input value is at least \$100. A shortfall occurs when the observed vehicle family returns more than one basis point less output than its feasible rival. Conditional quantiles use routes with such a shortfall; input-value-weighted means use every route in the stated sample and assign zero below the one-basis-point threshold. Panel B retains routes at least 30 days after a pair first uses one vehicle family exclusively and on which the route keeps that incumbent family. Panel C measures age from the pair’s first observed supported date. Panel D uses the same 52,207 routes in its gross and gas-adjusted columns. Path gas is predicted from year, transaction-callee class, ordered venue sequence, leg count, and cross-venue status; the counterfactual holds the observed callee class and effective gas price fixed. Receipt gas includes router transfers and bookkeeping. Appendix Table 17 reports held-out accuracy and path-specific bounds. Private order flow and venues outside the quoted set remain outside the comparison.

6 Discussion and policy implications

We group stablecoins by their shared vehicle function while retaining the token that supplies each route. USDC, USDT, DAI, and other stablecoins can target the same unit of account while differing in issuer, backing, redemption, peg risk, and local pool liquidity. Peg stress can reallocate trading across issuers (Liu et al., 2026; Anadu et al., 2025). The family-level result measures the rise of tokenised-dollar intermediation; the issuer-level estimates show how that role is allocated within the family. Stable-to-stable endpoint pairs identify this distinction directly because USDT supplies most of their increase in routed value.

A one-for-one peg aligns prices while preserving currency identity. The Bahamian dollar, for example, is conventionally pegged at one Bahamian dollar per U.S. dollar and remains a separately issued currency with its own monetary and exchange-control arrangements (International Monetary Fund, 2025). The same distinction disciplines our stablecoin aggregation. A currency-equivalent measure would consolidate dollar-pegged instruments by unit of account; our route measure records competition among separately issued dollar instruments and between that family and WETH.

This distinction may also affect liquidity supply. While two stablecoins hold their pegs, their relative price typically varies less than either token’s price against WETH. In a constant-product pool, smaller relative-price movements imply smaller loss versus holding, commonly called impermanent or divergence loss. That risk profile may make stablecoin–stablecoin links attractive to liquidity providers and help tokenised dollars build dense cross-issuer markets. Lower prior endpoint–vehicle risk predicts deeper constant-product bridge capital at the measurement date and 30 days later (Appendix Table 29, panel A, columns 1–4). Stablecoin bridges nevertheless have higher median relative volatility than WETH bridges and carry the lower risk in fewer than one third of comparable pair-months (panel B). Divergence risk helps explain local capital placement but leaves the aggregate stablecoin rotation unexplained. Impermanent loss remains distinct from loss-versus-rebalancing, the running adverse-selection cost of stale AMM prices (Milionis et al., 2022).

A newly issued stablecoin also gains convertibility by linking to incumbent dollar tokens, so issuer-sponsored or affiliated-market-maker capital may seed cross-issuer pools alongside independent liquidity supply. Deposited capital reveals the pool stock; provider identities, funding relationships, entry prices, fee income, hedges, and expectations determine profitability. We therefore use these channels as economic interpretations of pool formation.

Vehicle use connects otherwise unrelated pairs to the liquidity and issuer conditions of the asset in the middle. This concentration can lower trading costs through deeper two-leg markets while transmitting a disruption in those markets across many endpoint pairs. It is a network form of commonality in liquidity (Chordia et al., 2000). The pair-entry estimates show persistence after those links form. These are market-structure implications of observed route allocation. Welfare effects require additional evidence on user surplus, provider returns, and the incidence of stablecoin risk.

Payment-system design faces the same extensive margin. Nexus assigns foreign-exchange con-

version to a provider, and Rialto uses a third currency when direct exchange is unavailable ([Bank for International Settlements Innovation Hub, 2024, 2025](#)). The route evidence suggests that access policy affects more than the price quoted inside an existing relationship. It also affects which two-sided markets form early enough and deeply enough to become the bridge for new relationships.

7 Conclusion

Stablecoin vehicle dominance rises sharply on decentralised exchanges. Between matched January–June periods in 2024 and 2026, the equal-day stablecoin share among native and stable intermediaries rises from 16.9% to 42.1% by route count and from 32.7% to 76.5% by routed value (Appendix Table 21). In the pooled decomposition, movements toward and away from stablecoins offset inside continuing pairs, while activity shifting across continuing pairs contributes +8.4 pp by route count. Pairs first observed after 2024 H1 contribute +20.1 pp before reactivation, vehicle-role turnover, and exit reduce the period-specific net to +17.8 pp (Table 2, panel A). First-observed pair entry contributes +21.9 pp by routed value (panel B). Longer routes show the same movement.

The vehicle used at pair entry predicts later use after the entry day is removed from every outcome (Table 3, panel A). When a stablecoin first has at least \$10,000 of prior-day full-range capital on both required legs, 38.3% of pairs adopt it within 30 days and 47.0% within 120 days (Table 4, panel A). At the first sampled exact contest, the entry family carries 83.1% of routes; current output predicts retention, while the separate prior-capital estimate is imprecise (Table 5, panels A and B). Across established pairs, current output and prior capital both predict retention (Table 6, columns 3 and 4).

Price-rank crossings reveal how that retention breaks. Incumbent share falls 29.0 pp more at the crossing than during the same event’s earlier movement (Table 7, panel B, column 4). A 10 pp larger challenger share of prior weak-leg capital predicts 3.70 pp greater probability that the new price lead lasts one month (panel B, column 2). Among routes that retain a mature incumbent, the input-value-weighted gross-output shortfall is 3.3 bp (Table 8, panel B). Exact-price reversals move route flow; prior two-leg capital predicts whether the new lead persists.

Aggregate dominance combines use within continuing relationships with the vehicle identity of new ones. The large stablecoin rotation comes mainly from pair entry and where trading grows across continuing pairs. Inside established pairs, price competitiveness associated with two-sided depth maintains vehicle dominance, while exact-price rank reversals reallocate flow. Currency competition therefore operates when new relationships form their supporting markets and later when prices challenge the vehicle those markets established.

A From contract events to economic routes

Route construction begins with the contract and event format of each exchange. We identify pools from the protocol’s factory or registry, map contract addresses to tokens, and convert raw integer amounts using the token precision that applied at the time of the trade. Contract addresses define token identity because unrelated tokens can share a symbol and the same economic asset can appear in native and wrapped form. ETH and WETH are consolidated only after this address-level reconstruction; every other token remains distinct unless a stated economic classification combines it. Within each transaction, pool-swap legs are ordered by block, transaction, and log position. The resulting token flows reveal the exchange that occurred. A single connected chain gives one route and therefore one exchange between its pair endpoints; parallel chains identify an order split, and disconnected event groups remain separate. Because one transaction can contain unrelated calls as well as trades across several exchanges, this procedure also prevents two simultaneous routes from being mistaken for one longer route. Pool identities enter reconstruction; the unified route panel retains the ordered venue and token sequence together with the pair. Sender and router labels are retained separately.

Disconnected event groups become more common late in the sample. They comprise 0.8% of reconstructed components in 2024 and 6.2% in 2026. In 2026 the rate is 9.4% among components touching Uniswap v4 and 4.7% among other components. Table 9 uses the internally connected token flow of each group as a separate route. This alternative preserves the principal economic result: the 2026 stablecoin share is 43.7% by count and 77.3% by value, both slightly above the connected-transaction estimates.

Table 9: Disconnected transaction components and vehicle dominance

Sample boundary or estimate	2024	2025	2026
<i>Panel A. Share of reconstructed components excluded</i>			
All components	0.8%	3.7%	6.2%
Components touching Uniswap v4		11.6%	9.4%
Other components		2.7%	4.7%
<i>Panel B. Stablecoin share among native and stable two-leg vehicles</i>			
	2024	2026	Change [pp] (s.e.)
Connected routes, count	16.9%	42.3%	+25.4 (1.05)
Connected routes, value (20% agreement)	32.7%	76.5%	+43.9 (2.02)
Each valid component, count	16.9%	43.7%	+26.8 (1.06)
Each valid component, value (20% agreement)	32.9%	77.3%	+44.4 (1.93)

Note: Panel A reports the share of reconstructed components belonging to transactions with more than one disconnected token-flow group. Uniswap v4 begins during 2025, so its 2024 cell is empty. Panel B compares the principal connected-transaction rule with an alternative that treats each internally well-formed component as a separate route. Stablecoin shares use native and stable two-leg vehicles as the denominator and equal-weight matched January through June dates in 2024 and 2026. Value estimates require source, intermediary, and destination values to agree within 20%. Standard errors use a Newey–West covariance with a 30-day Bartlett bandwidth. The alternative assigns no unobserved handoff between components.

For three exchange families, full-day Ethereum-log reconciliations cover 206 venue-days across

74 sampled dates. We compare every provider swap with logs queried independently from the chain. The comparison can add a chain event absent from the provider data, remove a provider event absent from the chain, or replace its payload or log position. We then rerun the same route construction on the 50 sampled dates with at least one swap correction. Table 10 reports both the event match and its consequences for the economic route. Available indexed records omit 35 historical auxiliary venue-days; the table note reports their coverage and the log-order collision that sets the comparison’s boundary. The measured scope is the sampled dates and the connected pool sequence observed in each transaction.

Table 10: Ethereum-log checks and route-construction consequences

Panel A. Swap events against full-day Ethereum logs						
Venue	Days	Provider swaps	Provider only	Chain only	Precision [%]	Recall [%]
Uniswap v2	74	8,837,588	0	48	100.0000	99.9995
SushiSwap v2	70	599,721	0	0	100.0000	100.0000
Uniswap v3	62	4,444,960	7	2,957	99.9998	99.9335
Panel B. Assignments in transactions touched by a swap correction						
Assignment	Linked transactions			Changed	Unchanged [%]	
Ordered endpoint pair	4,881			363	92.563	
Intermediary token set	4,881			363	92.563	
Intermediary asset-type set	4,881			363	92.563	
Leg count	4,881			376	92.297	
Exact two-leg inclusion	4,881			206	95.780	
Panel C. Stablecoin shares on corrected route dates						
Measure	Provider rows [%]		Corrected rows [%]		Change [pp]	
Route count	24.446		24.438		-0.008	
Routed value	47.836		47.820		-0.016	
Panel D. January–June sampled decomposition, 2024 to 2026 [pp]						
	Route count			Routed value		
Component	Provider	Corrected	Change	Provider	Corrected	Change
Stablecoin-share change	+25.920	+25.921	+0.001	+40.566	+40.566	-0.000
Within continuing pairs	+0.078	+0.078	+0.000	+1.484	+1.484	-0.000
Across continuing pairs	+6.888	+6.888	+0.000	+14.635	+14.635	-0.000
Continuing-pair support mass	+0.895	+0.895	+0.000	-2.717	-2.717	-0.000
Pairs observed in one window	+18.059	+18.060	+0.001	+27.164	+27.164	+0.000

Note: Panel A uses full-day venue ledgers for 206 venue-days across 74 sampled dates. Provider-only events are provider rows absent from the corresponding Ethereum logs; chain-only events are Ethereum-log events absent from the provider rows. Panels B and C rerun the route construction on the 50 dates with at least one swap correction. Panel D repeats the pair decomposition on the six common January–June sampled dates in each endpoint year. Available indexed records omit 35 historical auxiliary venue-days. Auxiliary log-order corrections would collide with another provider-ordered leg in 2 transactions; together they contain 47 legs and 2 reconstructed routes. The comparison validates the sampled dates; transaction-trace recall beyond them remains unmeasured.

A complete token sequence is sufficient for the route-count sample. Value weighting additionally requires coherent amounts on every leg and a common dollar valuation. Routes without a coherent dollar valuation remain in counts and drop from value weighting. Appendix F gives the separate exclusion for self-returning routes.

A hypothetical AAVE–WETH–USDC–UNI route makes the units concrete. Suppose one AAVE is worth \$100, one WETH \$2,500, one USDC \$1, and one UNI \$10. If one AAVE returns 0.04 WETH, the second leg returns 100 USDC, and the third returns 10 UNI, then $q = \$100$, $Q_r = 10$ UNI, and $O_r = \$100$. Each leg carries \$100 on both sides. The route value ν_r is also \$100 because it averages the \$100 source and destination values. In the data, the same calculation uses transaction-time prices and allows limited measurement error under the value-agreement rule.

Historical precision is unverified for 13 tokens traded in 22 constant-product pools. Reserve-reconstruction analyses omit those pools. An upper-bound check removes every route involving any of the 13 tokens: the resulting bound is 5,950 routes and \$10.91 million of route value. Only 1,036 of those routes use one of the affected tokens as an intermediary, representing \$1.284 million, or 0.00043% of value on routes whose legs can all be valued. All 13 tokens lie outside the native and stablecoin groups used in the main comparison. The exclusion affects exact reserve reconstruction; the route-composition result is unchanged.

Table 11: Protocol architecture and native-asset pairing

Measure	Estimate
<i>Panel A. Uniswap v1: ETH intermediation required</i>	
Reconstructed token-to-token routes [count]	217,003
<i>Panel B. Uniswap v2: arbitrary token pools allowed</i>	
Single-leg trades executed in WETH pools, 2026 [%]	95.5
Token combinations first traded with WETH, 2026 [%]	97.9

Note: Panel A reports connected Uniswap v1 token-to-token routes reconstructed from the exchange registry and matched ETH legs. The protocol allowed only ETH–token pools, so each such route crossed ETH. Panel B reports Uniswap v2 activity during January through June 2026. A single-leg trade uses a WETH pool when either pool token is WETH. A token combination first traded with WETH when either token in the newly active combination is WETH. V2 allowed arbitrary ERC-20 token pools throughout this period.

B Pricing models and validation

The counterfactual comparison requires an exact-input quote from each pool’s state and the proposed input amount. Because pool families use different invariants and arithmetic, we validate their pricing models separately. For selected executed swaps, we reconstruct the state immediately beforehand and compare predicted with realised output. Curve’s amplification coefficient and Balancer’s weight ratio are unavailable in the historical subgraph records, so we estimate them on one set of trades and evaluate the resulting quotes on held-out trades. A route enters the cost comparison only when the relevant pool states and pricing models are available before the

transaction.

For a route r , let $e(r)$ record its block, transaction position, and first pool-event position; $e(r)^-$ is the state immediately before that transaction begins. The route has an ordered pool sequence $(p_1, \dots, p_{m(r)})$ and token sequence $(k_0, \dots, k_{m(r)})$, where $k_0 = i$ is the input and $k_{m(r)} = o$ is the output. The dollar input notional q corresponds to $\Delta_i(q, e) = q/P_{i,e}$ units of the input token. Let χ_{p,e^-} denote pool p 's pretrade state and let $Q_p(i \rightarrow j, \Delta_i; \chi_{p,e^-})$ denote its exact-input output in token j . Composing the pool quotes gives the route's token output and dollar value,

$$\begin{aligned} \Delta_{k_h} &= Q_{p_h}(k_{h-1} \rightarrow k_h, \Delta_{k_{h-1}}; \chi_{p_h, e(r)^-}), & h &= 1, \dots, m(r), \\ Q_r(q, e(r)) &= \Delta_o, & O_r(q, e(r)) &= P_{o, e(r)} Q_r(q, e(r)). \end{aligned} \quad (6)$$

Pool invariants take token quantities as arguments, so the dollar notional q and token input Δ_i remain distinct. Equation (6) describes an executed route and, after reconstructing the same pretrade state, prices a feasible alternative path. The subsections below specify Q_p for each pool family.

B.1 Constant-product pools

Uniswap v2 and SushiSwap v2 price the product of reserves (Lehar and Parlour, 2025). Let R_{i,p,e^-} and R_{o,p,e^-} denote pool p 's input- and output-token reserves immediately before transaction e , Δ_i the token input, f_p the fee rate, and $\varphi_p = 1 - f_p$ the fraction of input retained for pricing. An exact-input swap moves the pool along

$$(R_{i,p,e^-} + \varphi_p \Delta_i)(R_{o,p,e^-} - \Delta_o) = R_{i,p,e^-} R_{o,p,e^-},$$

which inverts in closed form to

$$\Delta_o = \frac{\varphi_p \Delta_i R_{o,p,e^-}}{R_{i,p,e^-} + \varphi_p \Delta_i}, \quad \varphi_p = \frac{997}{1000},$$

the 30 basis point fee both venues charge, applied on chain as the integer ratio above.⁵

A multi-leg route composes these one hop at a time, threading the output token of one pool into the next. The cost comparison evaluates the dollar outputs of the best direct and intermediated routes at the same state and dollar input notional,

$$10^4 \Delta C_{i,o,k,q,e}^D = 10^4 \frac{O_{i,o,q,e}^D - O_{i,o,k,q,e}^I}{O_{i,o,q,e}^D}.$$

Both outputs must be computed from one reconstructed pre-transaction state. A realised trade supplies the output of the chosen route alone; the direct and intermediary alternatives require reconstructed states for every pool in the comparison. Quoting each executed constant-product

⁵The SushiSwap v2 pool contract implements the same adjustment; see [UniswapV2Pair.sol](#).

swap against the state immediately before it gives a median absolute error of 0.0000% across 8,024 swaps; 96.7% of quotes are within 1% and 95.2% are within 0.01% of realised output.

B.2 Concentrated liquidity

Uniswap v3 distributes liquidity across price ranges, so executable depth depends on the active range and on the initialized ticks reached by a trade (Adams et al., 2021). Within one active range, let $\tilde{R}_{i,p,e^-}$ and $\tilde{R}_{o,p,e^-}$ denote the virtual reserves of the input and output assets. Liquidity L_{p,e^-} and the output-per-input marginal price ρ_{p,e^-} satisfy

$$\tilde{R}_{i,p,e^-} = \frac{L_{p,e^-}}{\sqrt{\rho_{p,e^-}}}, \quad \tilde{R}_{o,p,e^-} = L_{p,e^-} \sqrt{\rho_{p,e^-}}.$$

If f_p is the pool fee and $\varphi_p = 1 - f_p$, an exact input Δ_i returns

$$\Delta_o = \frac{\varphi_p \Delta_i \tilde{R}_{o,p,e^-}}{\tilde{R}_{i,p,e^-} + \varphi_p \Delta_i}$$

until the trade reaches the next initialized tick. Range endpoints lie on a geometric grid, and crossing tick j updates active liquidity by its signed change,

$$\sqrt{\rho(j)} = 1.0001^{j/2}, \quad L_{p,e^-} \mapsto L_{p,e^-} \pm \Delta L_j,$$

with the sign determined by the direction of the trade. The calculation then continues from the new active range until the input is exhausted. The quoter uses the contract’s integer Q96 arithmetic, rounding input amounts up and output amounts down.

The active tick map is accumulated from every mint and burn strictly before the validation day, after which same-day liquidity changes and swaps are replayed together by block and log index. Consecutive swaps in one pool supply the price state: the earlier event leaves `sqrPriceX96` and `tick` immediately before the later swap, while the interleaved event stream supplies active liquidity. We report errors separately by input direction and tick crossing because those cases use different arithmetic branches. Table 12 applies this validation to Uniswap v4. V4 retains concentrated-liquidity pricing for static-fee pools but implements it through a singleton and flash accounting.⁶ The validated static fee tiers include 0, 7, 8, and 20,000 hundredths of a basis point. Hook-bearing and dynamic-fee pools are excluded because the raw records do not identify their transaction-level cash flows and realised fees; Appendix E reports their share of v4 trading.

Every direction and tick-crossing row of Table 12 reproduces realised output within 0.001% in the Maximum column.

⁶See Adams et al., *Uniswap v4 Core*, Sections 1 and 3.

Table 12: Concentrated-liquidity quoter, by direction and tick crossing

Validation group	Swaps	Median	90th pct.	Maximum	Within 1%
token0 in, no tick crossed	2,290	0	0	9.98×10^{-4}	100.0%
token0 in, tick crossed	57	0	0	3.45×10^{-7}	100.0%
token1 in, no tick crossed	1,816	0	1.69×10^{-8}	1.48×10^{-4}	100.0%
token1 in, tick crossed	55	0	5.44×10^{-9}	8.74×10^{-9}	100.0%
Pooled	4,218	0	1.15×10^{-14}	9.98×10^{-4}	100.0%

Note: Absolute quote error in percent of realised output. The sample contains 4,218 realised Uniswap v4 swaps on three dates across 13 pools and nine token combinations, at static fee tiers 0, 7, 8, 100, 500, 10,000 and 20,000. Each quote uses liquidity events ordered strictly before the swap and the price state left by the preceding swap in the same pool.

B.3 StableSwap pools

Curve interpolates between a constant-sum curve, which is flat and charges no slippage, and the constant-product curve of Section B.1.⁷ For n tokens with normalized pre-transaction balances R_{j,p,e^-} , amplification A_p , and invariant D_p ,

$$A_p n^n \sum_j R_{j,p,e^-} + D_p = A_p D_p n^n + \frac{D_p^{n+1}}{n^n \prod_j R_{j,p,e^-}},$$

where $A_p \rightarrow 0$ recovers the constant product and $A_p \rightarrow \infty$ recovers the constant sum. The contract solves D_p by Newton iteration and then solves the same invariant for the post-trade output balance $R'_{o,p,e}$ after R_{i,p,e^-} is increased by Δ_i . Defining $D_\Pi = D_p \prod_j D_p / (n R_{j,p,e^-})$, the two loops are

$$D_p \leftarrow \frac{(A_p n \sum_j R_{j,p,e^-} + n D_\Pi) D_p}{(A_p n - 1) D_p + (n + 1) D_\Pi}, \quad R'_{o,p,e} \leftarrow \frac{(R'_{o,p,e})^2 + c}{2 R'_{o,p,e} + b - D_p},$$

with b and c formed from the balances of the tokens the swap does not touch. Balances are rescaled to eighteen decimals before either loop runs, because the invariant treats its tokens as interchangeable at par and a six-decimal stablecoin read on an eighteen-decimal scale is off by a factor of 10^{12} . Output is net of the pool fee,

$$\Delta_o = (R_{o,p,e^-} - R'_{o,p,e} - 1) \varphi_p,$$

carrying the contract's own off-by-one guard on the raw difference.

A_p is Curve-specific and the standardised subgraph schema drops it. We identify it from data because every other quantity in the invariant is observed and every Curve swap reports both amounts,

$$\hat{A}_p = \arg \min_{A_p} Q_{0.90} \left\{ \frac{|Q_p(i \rightarrow o, \Delta_i; \chi_{p,e^-}(A_p)) - \Delta_o|}{\Delta_o} \right\},$$

⁷See Egorov, *StableSwap—Efficient Mechanism for Stablecoin Liquidity*.

Table 13: StableSwap calibration and held-out quote error, by validation day

Day	Pools fitted	Pools excluded	Held-out trades	Median $\hat{A}_p$	Median error	Within 1%
2021-05-28	9	14	367	12,716	0.030%	100.0%
2022-09-12	11	16	262	20,493	0.029%	100.0%
2023-12-28	20	28	471	23,428	0.039%	98.9%
2025-04-13	42	50	1,207	46,851	0.036%	99.7%

Note: The sample contains four sampled days. $\hat{A}_p$ is reported in the module’s convention, which carries the on-chain precision factor of 100. Median error is the median absolute deviation of the quote from realised output on trades the calibration never saw.

fitted on the first half of a pool-day’s trades over a logarithmic grid with a ternary refinement, then evaluated on the trades not used for estimation. A pool-day is retained when the 90th percentile of fitting error is below 1%. Using the upper tail prevents a partial fit from passing: one misspecified pool produced a median fitting error below 0.1% but a 34% median error on the other trades because the invariant fit only part of its activity. Curve has also operated crypto-pools since 2021 that follow the CryptoSwap invariant. Applying StableSwap to these pools yields a best-fit A_p of 3 and a 36% median quote error, so pool-days whose observed trades are inconsistent with StableSwap remain outside this component exercise.

Table 13 reports the held-out fit by validation day.

Across four validation days, held-out median errors range from 0.029% to 0.039%.

B.4 Weighted pools

Balancer weighted pools use a weighted geometric-mean invariant,⁸

$$K_{w,p} = \prod_j R_{j,p,e^-}^{w_{j,p}}, \quad \sum_j w_{j,p} = 1,$$

which gives the two-token exact-input form

$$\Delta_o = R_{o,p,e^-} \left[1 - \left(\frac{R_{i,p,e^-}}{R_{i,p,e^-} + \varphi_p \Delta_i} \right)^{w_{i,p}/w_{o,p}} \right].$$

Only the ratio $w_{i,p}/w_{o,p}$ enters. An equal-weight Balancer pool and a constant-product pool with the same reserves therefore return the same quote. Balancer’s weighted-pool contract accepts inputs up to 30% of the input-token reserve; that contract limit governs whether a quote can execute, but it is not the empirical support rule.⁹ The counterfactual design applies the common 5% own-price-impact cutoff in Equation (8) to Balancer and every other pool family. The subgraph reports the weight ratio $\theta_p = w_{i,p}/w_{o,p}$ and the swap fee at its head block while balances come from a historical snapshot, so a pool that has repriced itself since is quoted with a current parameter against a

⁸See Martinelli and Mushegian, *A Non-Custodial Portfolio Manager, Liquidity Provider, and Price Sensor*.

⁹See the input-ratio bound in Balancer’s [WeightedMath.sol](#).

past state. Liquidity-bootstrapping pools move weights on a schedule by design and managed pools move them on demand, and a weighted pool’s owner can reset the fee at will. What such a pool produces is a small and nearly constant relative bias, which appears directly in the preceding validation: reported parameters reproduce trades in 2024 to a millionth of a basis point and trades in 2022 only to 0.19%. A gap of that size is a fee. The fee enters the quote through $\varphi_p \Delta_i$, so a fee wrong by a basis point moves output by about a basis point, whereas a mistake in the invariant would leave an error that scales with trade size.

We therefore treat both as unknowns to be recovered from the trades a pool actually executed, the move Curve’s amplification coefficient needed in Section B.3, and we keep a reported value only where it survives that test. Reconstructing a pool’s state from its own event history follows [Lehar and Parlour \(2025\)](#); what the weighted family adds is that two arguments of the pricing function are themselves missing from the historical record, so a reconstructed state alone does not deliver a quote. Every quantity in the weighted invariant other than θ_p and f_p is observed, and every Balancer swap reports both amounts, so with balances reconstructed either parameter is the only unknown. We order a pool-day’s trades and assign them alternately to a fitting set $\mathcal{O}_p$ and an evaluation set, and every error reported in Table 14 is measured on the evaluation set. For a trial parameter pair ϑ ,

$$\mathcal{E}_p(\vartheta) = Q_{0.90} \left\{ \frac{|Q_p(i \rightarrow o, \Delta_i; \chi_{p,e^-}(\vartheta)) - \Delta_o|}{\Delta_o} : e \in \mathcal{O}_p \right\}$$

is the fitting error, defined only when at least nine tenths of $\mathcal{O}_p$ returns a quote at all and undefined otherwise. The upper tail does the gating for the reason it does at Curve, and the coverage condition stops a parameter that prices one corner of a pool’s activity from standing for the pool.

Each token combination’s weight ratio is fitted on the trades crossing it in either direction, with a trade running from b to a scored at the reciprocal exponent, which is what the invariant implies and which doubles the trades available to pin one number,

$$\hat{\theta}_{ab,p} = \arg \min_{\theta} \mathcal{E}_p(\theta \text{ on } \mathcal{O}_p^{a \rightarrow b}, \theta^{-1} \text{ on } \mathcal{O}_p^{b \rightarrow a}),$$

searched over a logarithmic grid spanning Balancer’s own weight bounds and refined by ternary search in the log exponent, and attempted only for token combinations carrying at least four fitting trades, since one or two are fitted exactly by construction and then read as an exact fit. The fee $\hat{f}_p$ is recovered the same way over the fee range the vault permits. Reading is tried before fitting and the fee before the weights, so each tier adds at most one free scalar and a pool that reproduces its

trades on reported parameters is never handed a fitted one,

$$\hat{\vartheta}_p = \begin{cases} (\theta_p, f_p) & \text{if } \mathcal{E}_p(\theta_p, f_p) \leq \bar{e}, \\ (\theta_p, \hat{f}_p) & \text{else if } \mathcal{E}_p(\theta_p, \hat{f}_p) \leq \bar{e}, \\ (\hat{\theta}_{\cdot,p}, f_p) & \text{else if } \mathcal{E}_p(\hat{\theta}_{\cdot,p}, f_p) \leq \bar{e}, \\ \text{excluded} & \text{otherwise,} \end{cases} \quad \bar{e} = 0.1\%. \quad (7)$$

The label a pool carries in the source plays no part in this. A pool-day whose observed trades no tier reproduces is excluded whatever its `poolType` says, and the third tier further requires the fitted ratios to cover nine tenths of the fitting trades, so a pool-day is never accepted on the token combinations that happened to fit and then quoted on the rest.

The frequency with which the parameter search binds measures the sufficiency of reported pool state. Of the 256 pool-days accepted across the twelve validation days, 209 satisfy the error threshold on reported parameters with nothing identified, 43 need the swap fee alone, and 4 need per-token-combination weight ratios. Four fifths of the priced sample therefore carries no fitted parameter, and identification repairs a stale field in a minority of pool-days instead of supplying the invariant.

We set $\bar{e}$ at 0.1% instead of the 1% Curve uses because the achieved errors separate cleanly there. The threshold separates weighted pools, whose held-out errors range from 10^{-9} to 0.004%, from stable, composable-stable, and elliptic-curve pools, whose held-out errors range from 0.05% to 1.08% and have a 99th percentile of 12%. At a 1% threshold, fitted weight ratios would admit the second group. The 0.1% rule excludes it. Curve’s failed validation arose from an over-permissive parameter range; here, an over-permissive error threshold would produce the same mistake.

Across the twelve date rows of Table 14, the Backward column returns a median error of 0.0000% and the Pools priced / excluded column identifies 256 of 473 testable pool-days as weighted pools. The other pool-days carry a median 91.8% of tested swap volume and follow different Balancer pricing rules. This validates identified weighted pools; other Balancer families use different pricing rules.

Table 14: Weighted-pool quoter and reserve-timing alternatives

Day	Held-out trades	Backward	Flat	Opening	Pools priced / excluded
2021-04-25	4	4.7×10^{-9}	3.00%	3.94%	1 / 0
2021-09-29	273	2.5×10^{-8}	1.24%	0.80%	18 / 8
2022-03-09	524	1.6×10^{-4}	1.05%	2.66%	25 / 4
2022-08-16	701	1.1×10^{-9}	0.75%	1.04%	33 / 12
2023-01-24	832	2.0×10^{-10}	3.73%	4.71%	44 / 11
2023-07-03	301	7.1×10^{-10}	1.78%	3.32%	26 / 19
2023-12-11	185	2.8×10^{-9}	1.77%	2.91%	17 / 20
2024-05-20	265	3.8×10^{-10}	4.81%	5.17%	15 / 36
2024-10-27	142	1.8×10^{-10}	1.68%	2.24%	12 / 36
2025-04-06	248	2.1×10^{-10}	5.32%	9.48%	19 / 40
2025-09-13	501	2.5×10^{-4}	1.09%	1.71%	27 / 25
2026-02-21	582	8.7×10^{-9}	0.68%	1.90%	19 / 6

Note: Median absolute quote error in percent of realised output, on trades no fit ever saw. *Backward* reconstructs per-swap balances by rolling the day’s event sequence back from the closing snapshot, *Flat* keeps the snapshot balances constant across the day, and *Opening* treats the snapshot as the day’s opening state and rolls forward. The sample contains twelve days from 2021-04 to 2026-02.

C Reconstructing pool balances before each trade

Every counterfactual quote uses the pool state immediately before the trade, whereas some historical records store balances only at the end of an hour or day. We recover the earlier state by ordering every balance-changing event within the period and unwinding the sequence from the recorded closing balance. The exercises below validate that procedure on selected constant-product and weighted-pool observations. The cost comparison retains only routes whose pools can be reconstructed in this way.

Let $\mathbf{R}_p^{\text{end}}$ be pool p ’s stored period-end reserve vector and $\Delta_{p,e}$ the vector of net signed token amounts in event e . The state before event e is the stored value with every later event removed,

$$\mathbf{R}_{p,e}^{\text{pre}} = \mathbf{R}_p^{\text{end}} - \sum_{e' \geq e} \Delta_{p,e'} = \mathbf{R}_{p,e+1}^{\text{pre}} - \Delta_{p,e},$$

which is a single reverse pass over the period. The alternative reading takes the previous period’s stored value as the current period’s opening state and replays forward,

$$\tilde{\mathbf{R}}_{p,e}^{\text{pre}} = \mathbf{R}_{p,\text{prev}}^{\text{end}} + \sum_{e' < e} \Delta_{p,e'},$$

which accumulates every unrecorded balance change between the two stored points. Replaying forward returns a median absolute quote error of 11.8% against realised outputs, whereas unwinding backward returns 0.0000%.

Liquidity additions and withdrawals also move constant-product pool balances; the ordered

replay includes them alongside swaps. We unwind the sequence to the start of the period and compare the reconstructed opening balance with an independently observed reserve state,

$$\left\| \left(\mathbf{R}_{p,1}^{\text{pre}} - \mathbf{R}_{p,\text{prev}}^{\text{end}} \right) \oslash \mathbf{R}_{p,\text{prev}}^{\text{end}} \right\|_{\infty} < \tau,$$

Agreement establishes that the event history reproduces the observed reserves for the tested period. The state-dependent sample omits a period when a required event family is incomplete or the balances do not reconcile. Coverage is reported by exchange and year because missing state can be concentrated in actively managed or newly launched pools.

The same logic applies to weighted pools. We treat `poolSnapshots.amounts` as the closing balance and unwind joins, exits, and swaps in their recorded order. Table 14 compares this reconstruction with two alternatives that either hold the closing balance fixed or treat it as the opening state.

The Balancer replay includes joins and exits as well as swaps. Omitting those events assigns intraday liquidity changes to the pricing rule and changes the pool states available for the price comparison.

D Restricting hypothetical trades to observed size–depth combinations

Component-validation samples contain executed swaps and therefore pools deep enough to serve the observed trade. Hypothetical routes can demand much more output relative to pool depth, where the observed errors no longer describe pricing accuracy. Let $\tilde{Q}_{\ell}$ denote leg ℓ ’s output at the pool’s marginal rate. We include a hypothetical leg only when

$$\pi_{\ell}(\Delta_i, e^{-}) = \frac{\tilde{Q}_{\ell}(\Delta_i, e^{-}) - Q_{\ell}(\Delta_i, e^{-})}{\tilde{Q}_{\ell}(\Delta_i, e^{-})} \leq \bar{\pi}, \quad \bar{\pi} = 0.05. \quad (8)$$

The ceiling is calculated under the pricing rule of the pool family the leg would traverse; contract-specific limits add further restrictions.

For constant-product swaps, input relative to reserves provides a transparent size–depth comparison,

$$\iota_{i,p,e^{-}} = \frac{\Delta_i}{R_{i,p,e^{-}}},$$

the input amount as a fraction of the input-side reserve. Table 15 reports its distribution in the historical validation sample. The statistic helps calibrate what constitutes a large trade in a constant-product pool, but it does not replace the common own-price-impact rule for concentrated-liquidity, StableSwap, or weighted pools.

In the Pooled row of Table 15, five per cent of input reserves sits between the 90th percentile of 3.29% and the 99th of 14.86%; the same cutoff lies between those columns on seven of the eight date

Table 15: Size relative to depth in the validation population

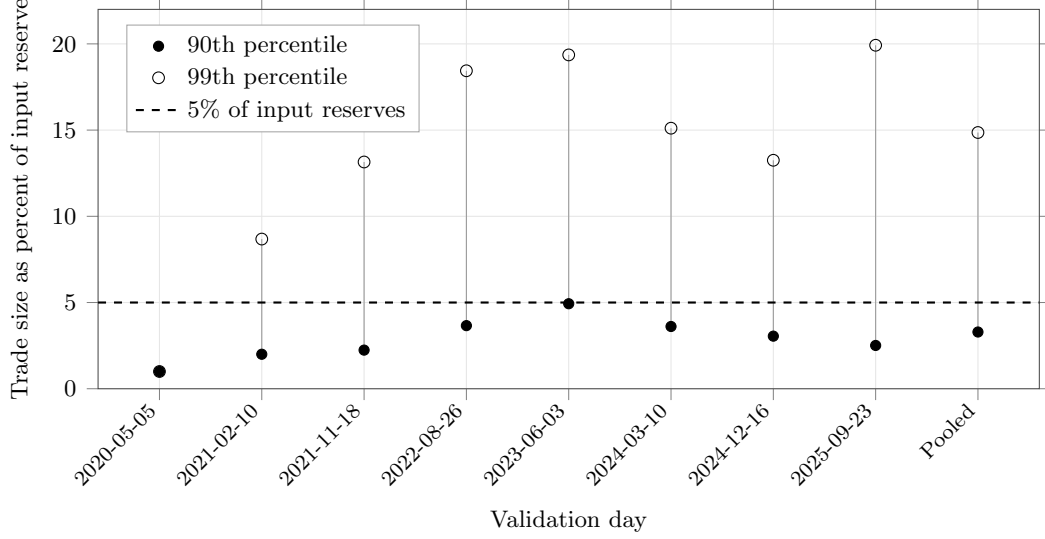
Day	Swaps	Median	90th pct.	99th pct.	99.9th pct.	Maximum
2020-05-05	2	1.00%	1.00%	1.00%	1.00%	1.00%
2021-02-10	141,804	0.19%	2.00%	8.68%	23.17%	99.83%
2021-11-18	107,059	0.10%	2.24%	13.15%	71.64%	100.00%
2022-08-26	79,180	0.48%	3.66%	18.44%	94.13%	100.00%
2023-06-03	159,551	0.85%	4.93%	19.36%	96.59%	100.00%
2024-03-10	215,060	0.41%	3.61%	15.11%	57.81%	100.00%
2024-12-16	132,508	0.31%	3.05%	13.25%	71.93%	100.00%
2025-09-23	97,106	0.24%	2.51%	19.92%	99.11%	100.00%
Pooled	932,270	0.34%	3.29%	14.86%	78.33%	100.00%

Note: Trade size as a percentage of the input-side reserve of the pool traversed, over the constant-product swaps the quoters were validated against. The sample contains eight dates from 2020-05-05 through 2025-09-23.

rows. Figure 2 shows the daily distributions. This calculation locates the cutoff in the empirical size distribution. The empirical rule remains a 5% own-price-impact cutoff computed separately under each pool family’s pricing function.

Equation (8) applies the same 5% own-price-impact ceiling to every family. The component exercises establish pricing accuracy for realised swaps within their stated samples; they do not measure quote error on unexecuted routes.

Figure 2: Constant-product trade size relative to reserves



Note: The statistic is a leg’s input amount as a fraction of the input-side reserve in the constant-product pool it traverses. Points mark the 90th and 99th percentiles; the dashed line marks 5% of reserves. The two percentiles coincide on 2020-05-05. The sample contains 932,270 realised swaps on eight dates from 2020-05-05 through 2025-09-23.

E Coverage of the price comparison

Realised routes identify the assets that carried each trade. The price frontier holds the source i , destination o , dollar input q , and pre-transaction pool state e^- fixed. The best direct and two-leg outputs through intermediary k are

$$O_{i,o,q,e}^D = P_{o,e} \max_{p \in \mathcal{P}_{i,o,e^-}} Q_p(i \rightarrow o, \Delta_i(q, e); \chi_{p,e^-}),$$

$$O_{i,o,k,q,e}^I = P_{o,e} \max_{p' \in \mathcal{P}_{k,o,e^-}} Q_{p'} \left(k \rightarrow o, \max_{p \in \mathcal{P}_{i,k,e^-}} Q_p(i \rightarrow k, \Delta_i(q, e); \chi_{p,e^-}); \chi_{p',e^-} \right).$$

The vehicle-route shortfall relative to direct execution is

$$\Delta C_{i,o,k,q,e}^D = \frac{O_{i,o,q,e}^D - O_{i,o,k,q,e}^I}{O_{i,o,q,e}^D},$$

reported in basis points as $10^4 \Delta C_{i,o,k,q,e}^D$. The comparison extends the realised-versus-optimised allocation benchmark of [Xi and Moallemi \(2026\)](#) from parallel pools connecting the same assets to sequential routes through an intermediary. Swaps and liquidity changes are replayed in on-chain order so every quote uses information available before the trade.

The opportunity set expands in three stages: alternative paths using the same vehicle and venue families as the observed route, paths using the same vehicle across Uniswap v2, SushiSwap v2, and Uniswap v3, and paths using any named public vehicle or the direct connection across

Table 16: Exact pre-transaction prices and vehicle identity

<i>Panel A. Coverage and quote validation</i>				
Linear routes in the exact venue set [%]				93.4
Mapped observed routes reproduced within 1 bp [%]				89.2
<i>Panel B. Nested opportunity set</i>				
	Realised route	Same vehicle, used venues	Same vehicle, all exact venues	Any named vehicle or direct
Routes with more than 1 bp higher output [%]	0.0	6.6	44.5	46.4
Stablecoin vehicle share [%]	19.5	19.5	19.5	18.4
Full-set minus realised stablecoin share [pp]			Route weighted	−1.16*** (0.19)
			Input-value weighted	−2.09*** (0.31)
Routes / dates				777,203 / 73

Note: The sample contains exact two-leg routes executed on Uniswap v2, SushiSwap v2, or Uniswap v3 on the fifteenth day of each month from June 2020 through June 2026. Source, destination, input amount, and pre-transaction pool state remain fixed. Each observed quote reproduces realised output within one basis point. Leg values must agree within 20%, input must be at least \$100, and every observed or hypothetical leg must remain below 5% own-price impact. The public vehicle set contains WETH, USDC, USDT, DAI, and WBTC; a realised vehicle outside that set remains eligible. A route changes vehicle only when gross output improves by more than one basis point. Standard errors in parentheses are clustered by date. P-values use Holm adjustment across route- and input-value-weighted comparisons in five reported samples. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The three exact venues contain 93.4% of linear routes over the sampled dates, and 89.2% of mapped observed routes reproduce within one basis point. Coverage falls as other exchanges expand late in the sample. The comparison is gross of gas and excludes private order flow. 98 routes imply more than twice realised output under standard invariant pricing; removing them leaves every displayed share unchanged at one decimal place, and they are excluded from the median-gain calculation.

those exchanges. Each observed two-leg route must reproduce realised output within one basis point. Leg values must agree within 20%, input value must be at least \$100, and every observed or hypothetical leg must satisfy Equation (8).

E.1 Realised gas burden

Receipt gas measures the fixed execution toll attached to adding a vehicle leg. In the executed-route sample, a stable-vehicle route in 2024 uses 94,290 more gas than the median direct route, 50.2% of the direct-route median. Evaluated at that year’s median gas price and WETH price, the extra toll is \$3.24, so a trade needs about \$32.4k of notional before the extra gas is below one basis point. In 2026 the median stable-vehicle route uses 233,640 more gas than the direct-route median, while the dollar toll falls to \$0.08 and the one-basis-point notional falls to \$762. At each stable-vehicle route’s

own gas price, median stable-vehicle notional falls from \$832 to \$82, the median fixed toll falls from 44.3 bp to 16.2 bp, and the share of routes with a toll below 25 basis points rises from 39.5% to 55.7%. The extra leg adds gas units but becomes small relative to most observed stable-vehicle notionals. These receipt summaries describe the realised fixed toll; the path-specific model below predicts the toll for both feasible vehicle paths.

For the path-specific comparison, we fit transaction gas on 40,186 receipt-measured transactions spanning April 2019 through June 2026. Predictors are year, transaction-callee class, ordered venue sequence, route-leg count, and cross-venue status. A deterministic holdout contains 7,520 transactions. Median absolute percentage error is 17.4%, compared with 34.5% for a leg-count-only model (Table 17, panel A). The central prediction matches 52,433 of 52,477 common-support routes to receipts and retains 52,207 with positive output net of predicted gas on both paths. Panel B applies the path-specific interquartile predictions in both directions. These bounds are wide for small trades, so the central adjustment enters the main comparison and the bounds remain here.

Table 17: Gas prediction and path-specific output bounds

<i>Panel A. Held-out prediction accuracy</i>						
Sample	Test routes	Actual gas	MAE	MAPE [%]	Legs-only MAPE [%]	IQR coverage [%]
All routes	7,520	290,864	49,762	17.4	34.5	45.0
Two-leg routes	4,860	289,920	49,651	17.4	35.1	46.0
<i>Panel B. Path-specific interquartile gas bounds</i>						
Input value	Lower-output route [%]			Weighted shortfall [bp]		
	Central	Favorable	Unfavorable	Central	Favorable	Unfavorable
All routes	13.7	6.2	30.1	7.65	6.31	11.85
\$100–999	14.7	4.2	41.7	22.01	12.01	104.67
\$1,000–9,999	13.0	6.5	24.1	9.51	6.46	18.48
\$10,000–99,999	13.1	10.5	16.5	9.21	8.04	10.83
\$100,000 and above	9.7	9.7	10.8	3.77	3.68	3.89

Note: Panel A uses a deterministic transaction-hash holdout from the 40,186-transaction receipt panel. Actual gas, median absolute error (MAE), and prediction features refer to total successful-transaction gas units, including router transfers and bookkeeping. The legs-only benchmark uses route-leg count alone. Interquartile-range (IQR) coverage is the share of realised receipt gas between the cell-specific 25th and 75th percentiles. Panel B subtracts path-specific gas from the exact stablecoin and WETH outputs using the executed transaction’s effective gas price and same-day WETH price. The central estimate uses cell medians. The favorable bound assigns the observed route its 25th-percentile gas and the rival path its 75th percentile; the unfavorable bound reverses them. The rival path retains the observed transaction-callee class. A shortfall starts above one basis point, and weighted means assign zero below that threshold.

E.2 Exchange and pool-family coverage

Omitting an exchange can change the best direct route, the best intermediated route, or both. Panel A of Table 18 describes the volume recorded in nine local source series, including Uniswap v1 and the available Fluid extracts. Volume in the Graph-indexed series is reconstructed from individual swaps using the smaller dollar value of the two legs. This avoids double counting in two series and removes implausible oracle values in thin pools: one Curve pool reports a \$456m day whose swaps sum to \$8, and a constant-product pool with \$255 of reserves reports \$1.36bn. Physical bounds on volume relative to reserves exclude between 0.10% and 0.24% of Graph-indexed pool-days per year. Uniswap v1 enters the full route panel but not the route-cost opportunity set because the price comparison requires a common pre-transaction state model. Fluid supplies volume on only 40 of the 400 sampled dates and no historical TVL field. Panel B uses DeFiLlama’s daily protocol breakdown to compare the selected exchange families with total Ethereum DEX volume. Their combined share declines from 98.5% in 2020 to 77.5% in 2026 H1 and is 87.5% over the displayed period. This external comparison describes the families’ market reach; it does not establish transaction-level or pretrade-state coverage for every deployment.

The Uniswap v4 pricing model covers static-fee pools without hooks. Pool-identity reconstruction leaves zero swaps with incomplete or invalid immutable pool attributes across 29,005,863 rows. Pools in this group account for 84.02% of swaps and 99.05% of reported value; the corresponding 2026 shares are 79.89% and 95.57%. Other pools use hooks, dynamic fees, or both, whose transaction-level cash flows are not identified from the raw event. These shares describe the reach of the pricing model. They do not establish the availability of every pre-transaction state.

Table 18: Exchange-source composition and Ethereum market coverage

<i>Panel A: shares within the nine retained source series</i>									
Year	Uni V1	Uni V2	Uni V3	Uni V4	Sushi V2	Sushi V3	Curve	Balancer	Fluid
2020	2.09	75.89	0.00	0.00	10.89	0.00	11.12	0.00	0.00
2021	0.02	31.27	37.97	0.00	17.31	0.00	12.25	1.18	0.00
2022	0.01	7.01	64.07	0.00	4.35	0.00	18.70	5.87	0.00
2023	0.01	9.26	66.81	0.00	1.21	0.03	13.88	8.80	0.00
2024	0.00	11.63	69.06	0.00	0.49	0.01	12.57	5.86	0.38
2025	0.01	4.21	53.10	19.53	0.28	0.01	9.68	1.49	11.70
2026	0.01	2.27	46.02	31.94	0.11	0.16	12.61	0.15	6.73
Pooled	0.05	14.58	54.54	5.30	5.73	0.02	13.37	3.82	2.61
<i>Panel B: selected exchange families as a share of total Ethereum DEX volume</i>									
Period	2020	2021	2022	2023	2024	2025	2026 H1	Pooled	
Share (%)	98.5	92.2	92.3	86.6	87.2	82.4	77.5	87.5	

Note: Panel A reports percentage shares of volume recorded in nine local source series. Its weekly calendar contains 400 dates; Fluid is observed on 40 (6 in 2024, 25 in 2025, and 9 in 2026) and supplies no TVL field. Panel B sums the protocol families represented in the route panel and divides by total Ethereum DEX volume in DeFiLlama’s daily breakdown; the numerator includes Uniswap v1–v4, SushiSwap v2 and v3, Curve, Balancer v2, and Fluid. DeFiLlama’s protocol taxonomy need not match the exact transaction-level deployments in Panel A. The 2026 cell ends on June 30, and the pooled cell sums 2020 through 2026 H1. See [DeFiLlama’s DEX-volume series](#).

Curve carries between 9.68% and 18.70% of observed source volume in every year it exists and ranks second in four years, making its coverage economically important. The StableSwap model fits 621 pool-days and excludes 588 whose 90th-percentile fitting error exceeds 1%. Across 28 sampled dates from 2020-02-11 to 2026-04-28, these exclusions account for 34.2% of the Curve volume assessed, or \$1.83bn of \$5.36bn. Another 2.75% of Curve volume occurs in pool-days with too few trades to estimate and evaluate the amplification coefficient separately. Lowering the minimum from eight trades to four changes annual excluded shares by at most 1.3 pp. Table 19 shows where these exclusions occur.

The exclusions are concentrated in routes involving the native asset. In Panel B of Table 19, the excluded share of native-leg volume exceeds the stable-leg share by at least 33 pp in every sample year and by 54 pp in the 2023 row. Tricrypto pools pairing WBTC and WETH with a stablecoin account for \$818m, or 44.6%, of the \$1.83bn excluded, including seven of the twelve largest excluded pool-days. These CryptoSwap pools concentrate the missing coverage where Curve supplies depth between ETH and stablecoins.

SushiSwap v3 is omitted from the route-cost comparison because its historical records lack the pool’s root price, in-range liquidity, and tick spacing. Its reserves are distributed across ticks, and the balance and weight fields exposed by the common schema do not describe that invariant. The exchange accounts for 0.016% of volume among the seven exchanges used for route costs over the full sample and 0.176% in 2026. Across fourteen sampled days, only 4.1% of its volume occurs on token combinations unavailable on another included exchange, never more than \$37,609 on one day.

Balancer accounts for 3.9% of measured venue volume over the full sample and 8.8% at its 2023

Table 19: Curve volume outside the StableSwap comparison

<i>Panel A: composition of the exclusion</i>					
Leg served	Pool-days	Volume	Of excluded	Of priced	Of that leg
Native	374	\$1.038bn	56.6%	15.7%	65.2%
Stable	119	\$0.752bn	41.0%	79.9%	21.1%
Other volatile	95	\$0.045bn	2.4%	4.4%	22.2%
<i>Panel B: excluded share of each leg’s volume, by year</i>					
Year	Native leg		Stable leg	Other volatile	
2021	85.49%		46.91%	15.23%	
2022	68.82%		30.21%	50.77%	
2023	59.11%		5.51%	56.60%	
2024	48.24%		0.94%	14.38%	
2025	47.47%		13.96%	10.52%	
2026	60.92%		9.04%	20.08%	

Note: A pool serves the *native* leg when it contains ETH or a staking derivative, the *stable* leg when every token is stable, including baskets and interest-bearing wrappers, and the *other volatile* leg otherwise. Stablecoin base-pool tokens are classified with the underlying stable assets. The sample contains 28 measured Curve pool-days from 11 February 2020 through 28 April 2026.

peak. Table 14 validates selected weighted pools. Linear and boosted pools remain outside the quoter. Of 29 sampled composable-stable pools, 23 reduce to two external tokens after removing the pool’s own share token, but pricing them also requires historical rate-provider states absent from the source. The priced Balancer sample is therefore confined to weighted pools whose historical state can be reconstructed and whose realised trades are reproduced at one of the tiers of Equation (7).

F Round-trip exclusion

A route whose first input token equals its last output token,

$$\mathbf{1}_{\{k_0=k_{m(r)}\}},$$

returns to its starting asset. We exclude these routes from the dominance measures. They include cyclic arbitrage and other self-returning activity; the endpoint rule alone leaves trader intent unidentified. Multi-trade arbitrage can be bundled within one transaction (Daian et al., 2020).

Table 20 reports the distribution across every day with at least 200 multi-leg routes.

At the median, round trips account for 12.8% of multi-leg routes by count and 21.0% by value. At the 95th percentile the shares are 23.0% and 51.1%, and the value-share maximum reaches 97.7% on 13 October 2023. The wider upper tail by value is consistent with large self-returning routes carrying disproportionate notional. Round trips are excluded before the count- and value-weighted dominance measures are formed.

Table 20: Round-trip share of multi-leg routing across all eligible days

Quantile across days	Share by count	Share by value
Minimum	0.00%	0.00%
25th percentile	9.66%	15.44%
Median	12.82%	20.96%
75th percentile	16.28%	27.99%
95th percentile	22.97%	51.07%
Maximum	33.09%	97.73%
2019 median	0.00%	0.00%
2020 median	12.85%	12.77%
2021 median	13.39%	23.95%
2022 median	14.17%	25.82%
2023 median	9.18%	17.07%
2024 median	10.38%	19.83%
2025 median	16.21%	30.47%
2026 median	21.21%	20.08%

Note: Round trips as a share of multi-leg routes. The sample contains all 2,449 eligible dates from 15 May 2019 through 30 June 2026 and 359,631,951 routes, of which 68,927,053 are multi-leg. Dates with fewer than 200 multi-leg routes are omitted.

G Additional composition and network results

For route r , let $\mathcal{M}(r)$ denote the assets between its source and destination and define $Z_{r,k} = \mathbf{1}_{\{k \in \mathcal{M}(r)\}}$. Let $a(k)$ assign each token to an economic asset group and set $Z_{r,g} = \sum_{k:a(k)=g} Z_{r,k}$ for $g \in \{\text{native}, \text{stable}\}$. On an exact two-leg route, $Z_{r,g}$ is an indicator. Let $\mathcal{R}_t^{(2),NS}$ contain the exact two-leg routes on day t whose intermediary is native or stable, and let $\mathcal{R}_t^{(2),NS,V}$ retain the routes whose source, intermediary, and destination values agree within 20%. The count- and dollar-weighted shares are

$$S_{g,t}^N = \frac{\sum_{r \in \mathcal{R}_t^{(2),NS}} Z_{r,g}}{|\mathcal{R}_t^{(2),NS}|}, \quad S_{g,t}^V = \frac{\sum_{r \in \mathcal{R}_t^{(2),NS,V}} \nu_r Z_{r,g}}{\sum_{r \in \mathcal{R}_t^{(2),NS,V}} \nu_r}, \quad (9)$$

where ν_r is the arithmetic mean of the route’s total dollar value at its observed source and destination. The value sample also requires each intermediary’s incoming and outgoing dollar values to have a ratio between 0.8 and 1.2. The native and stable groups exhaust the denominator in Equation (9).

Let W_y be the share of native-plus-stable activity in continuing pairs, $\omega_{p,y}$ pair p ’s weight within that activity, and $s_{p,y}$ its stablecoin share. Continuing pairs have stablecoin share $S_{C,y} = \sum_p \omega_{p,y} s_{p,y}$, while period-specific pairs have share $S_{E,y}$. Total stablecoin share is $S_y = W_y S_{C,y} +$

Table 21: Stablecoin share among native and stable intermediaries

Stablecoin share among native and stable intermediaries	2024	2026	Change [pp] (s.e.)
Route count	16.9%	42.1%	+25.2 (1.04)
Dollar-weighted routes (20% agreement)	32.7%	76.5%	+43.8 (2.01)

Note: The denominator contains exact two-leg routes whose sole intermediary is native or stable. Daily shares receive equal weight over matched January–June dates in 2024 and 2026. Dollar-weighted estimates require source, intermediary, and destination values to agree within 20%. Standard errors use a Newey–West covariance with a 30-day Bartlett bandwidth.

Table 22: Alternative accounting for the route-count rotation

Component	Estimate [pp]
Pairs entering or leaving the sample	+10.0
Pairs gaining or losing a vehicle route	−0.4
Market activity shifting across continuing pairs	+7.7
Change in how often continuing pairs use a vehicle	+6.9
Stablecoin share within continuing vehicle-using pairs	+1.3
Total route-count change	+25.5

Note: The table factors the pooled route-count change into pair support, vehicle-route support, activity reallocation, vehicle incidence, and stablecoin share within continuing vehicle-using pairs. It is an alternative accounting of the same total reported in Table 2, panel A; its components use a different mass and are not added to the main decomposition.

$(1 - W_y)S_{E,y}$. Writing $\Delta x = x_{2026} - x_{2024}$ and $\bar{x} = (x_{2024} + x_{2026})/2$ gives

$$\begin{aligned}
\Delta S = & \underbrace{\bar{W} \sum_p \bar{\omega}_p \Delta s_p}_{\text{switching within continuing pairs}} + \underbrace{\bar{W} \sum_p \bar{s}_p \Delta \omega_p}_{\text{activity shifting across pairs}} \\
& + \underbrace{(\bar{S}_C - \bar{S}_E) \Delta W}_{\text{continuing-pair weight}} + \underbrace{(1 - \bar{W}) \Delta S_E}_{\text{period-specific pairs}}.
\end{aligned} \tag{10}$$

The midpoint averages hold the companion quantity fixed while each component changes; the four terms add exactly to the aggregate change.

For route-count or routed-value stablecoin share $S_{pds,y}^{(m)}$, the matched comparison is

$$S_{pds,y}^{(m)} = \alpha_{pds}^{(m)} + \beta^{(m)} \mathbf{1}_{\{y=2026\}} + \varepsilon_{pds,y}^{(m)}, \quad m \in \{N, V\}, \tag{11}$$

where d is the month-day and s the realised route class. Observations are weighted by native-plus-stable route count or supported routed value, and standard errors cluster by pair and date.

Appendix Table 24 separates endpoint composition from intermediary identity inside stable-to-stable routes.

The full route panel supports two complementary measures. Intermediary-position share allocates all observed intermediary work across currencies. Route-participation share records the fraction of routes containing a currency and can sum above one across currencies because a longer

Table 23: Time-window and endpoint restrictions on the stablecoin rotation

<i>Panel A. Adjacent January–June comparisons, route count</i>							
H1 comparison	Total	Within pair	Across-pair shift	Support shift	Pair entry/exit		
2019–2020	+7.3	+0.0	+0.0	+3.0	+4.2		
2020–2021	+11.6	+0.5	+11.3	−6.4	+6.2		
2021–2022	+10.3	+0.5	+1.8	−1.5	+9.5		
2022–2023	−6.4	−0.4	−1.7	−4.5	+0.1		
2023–2024	−5.0	−0.4	+1.6	−6.8	+0.5		
2024–2025	+8.0	−0.1	+2.9	−2.1	+7.4		
2025–2026	+17.4	+1.0	+5.0	−1.1	+12.4		
<i>Panel B. Neither endpoint is WETH or a stablecoin</i>							
	2024 H1	2026 H1	Total	Within pair	Across-pair shift	Support shift	Pair entry/exit
Route count	1.1	9.0	+8.0	+0.2	+0.6	−0.0	+7.2
Supported value	0.9	39.1	+38.1	−0.3	+1.5	+3.1	+33.8

Note: Both panels apply the exact decomposition in Equation (10); endpoint-year shares are percentages, and all changes and components are in pp. Panel A uses equal-weight matched January–June dates in each adjacent pair of years. Panel B compares matched January–June dates in 2024 and 2026 after removing every pair with WETH or a stablecoin at either endpoint. Supported value requires source, intermediary, and destination values to agree within 20%. The across-pair component changes weights among continuing pairs, the support shift changes the aggregate weight on continuing pairs, and pair entry/exit is net of pairs present in only one endpoint year. Components use unrounded values and sum to the total change.

Table 24: Endpoint direction and stablecoin intermediary identity

	Route count [pp]	Routed value [pp]
<i>Panel A. Contribution by endpoint direction</i>		
All other pairs	+15.2	+20.6
One native, one stable endpoint	+9.2	+10.1
Two stable endpoints	+1.0	+13.2
Total stablecoin-share change	+25.4	+43.9
<i>Panel B. Stablecoin identity within two-stable-endpoint pairs</i>		
USDT	+0.6	+13.7
USDC, DAI, and other stablecoins	+0.4	−0.5
All stablecoin intermediaries	+1.0	+13.2
Top-decile-trimmed daily mean	−	+12.3
Pooled route-value weighting	−	+12.7

Note: Each entry is a contribution to the equal-day stablecoin-share change between matched January–June dates in 2024 and 2026. Panel A partitions the full native-or-stable intermediary sample by pair endpoints. Panel B partitions the stablecoin contribution within pairs whose two endpoints are stablecoins; its first two rows sum to the third. Routed value requires source, intermediary, and destination values to agree within 20%. The trimmed row removes the largest decile of daily contributions in each year. The pooled row weights routes by supported value across the matched dates. Components use unrounded values.

Table 25: All-route participation and trading-network position

<i>Panel A. All complete route lengths</i>				
Year	Currency family	Intermediary-position share [%]	Route-participation share [%]	Routes using family
2020	Native	71.7	78.2	2,915,214
	Stablecoin	25.4	26.0	970,851
2024	Native	75.3	79.5	6,894,487
	Stablecoin	17.2	17.6	1,523,281
2026 H1	Native	42.1	50.2	2,024,310
	Stablecoin	41.8	46.0	1,855,019
<i>Panel B. Named currencies in the trading graph</i>				
Currency	Year	Betweenness (rank)	Intermediary-position share [%]	Route-participation share [%]
WETH	2020	0.965 (1)	70.7	77.2
	2024	0.988 (1)	76.2	80.5
	2026 H1	0.925 (1)	42.4	50.7
USDC	2020	0.017 (4)	10.7	11.7
	2024	0.045 (2)	8.1	8.6
	2026 H1	0.107 (3)	21.1	25.1
USDT	2020	0.096 (2)	10.7	11.7
	2024	0.027 (3)	6.8	7.1
	2026 H1	0.113 (2)	16.2	19.3

Note: Panel A uses all coherent complete routes that do not return to their starting asset. Intermediary-position shares use all intermediary positions as the denominator. Route-participation shares use complete routes as the denominator and may sum above 100% across currency families. The 2020 and 2024 rows use full calendar years; 2026 H1 covers January through June. Panel B builds one undirected trading graph for each year from legs observed on the fifteenth day of each month. An edge records at least one leg in either direction during the sampled year. Betweenness is unweighted and approximated from 256 source nodes with a fixed seed. Realised-use shares in Panel B use the same sampled dates and retain all route lengths. Ranks are shown in parentheses.

route may contain several intermediaries. Panel A of Table 25 reports both measures for the native and stablecoin families. Panel B compares realised use with each named currency’s position in the trading graph.

Table 25, panel A, shows the same stablecoin rotation under both all-route normalisations. Panel B shows a different movement in graph position. WETH remains the first-ranked betweenness node in every year, even as USDC and USDT gain realised intermediary use. The result separates the existence of leg connections from the depth and execution conditions that determine route allocation.

The issuer split in Appendix Table 24, panel B, keeps the token-level result beside the endpoint composition it explains. Together with Table 25, these exhibits show that the rotation spans longer routes and separately issued stablecoins even though WETH remains the graph’s most central connector.

Table 26: Post-entry persistence among pairs with greater entry activity

<i>Stablecoin route share among entrants with greater first-day activity</i>				
	Days 1–30		Days 31–120	
Minimum first-day routes	5	10	5	10
Entry stablecoin share	+9.70***	+9.65***	+9.31***	+9.35***
[effect pp per 10 pp]	(0.19)	(0.32)	(0.44)	(0.60)
Observations (retrading pairs)	1,629	671	885	452
Eligible entrants	2,192	775	2,192	775
Entry-date clusters	115	93	100	80
Pair controls and equal-pair weights	Yes			

Note: Each column regresses the stablecoin share in the stated later window on the stablecoin share observed on the entry day. The sample requires complete follow-up through day 120 and subsequent trading in the stated window; the entry day is excluded from every later outcome. All columns control for cohort year, endpoint class, log entry routes, direct-route share, and longer-route share and use equal-pair weights. Coefficients and entry-date-clustered CR1 standard errors in parentheses report the outcome change in pp for a 10 pp higher entry share. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

H Secondary route and liquidity comparisons

We use the following estimates to interpret the main results: entry-activity restrictions, the future-persistence version of bridge support, the composition of capital growth before first use, and relative-price risk. Pool states and first-use dates are equilibrium outcomes; the estimates describe how route use and liquidity line up.

H.1 Persistence among pairs with greater entry activity

Table 26 repeats the post-entry stablecoin-share regressions after requiring at least five or ten routes on the entry day. The estimates remain close to one-for-one at both horizons. The main persistence result is also present among entrants with a more precisely measured initial allocation.

H.2 Capital composition and later adjustment

The main bridge event uses only prior-calendar capital. Table 27 gives the complementary event dated by subsequent persistence: positive prior-calendar capital on both legs during at least 24 of the next 30 days. It reproduces the increase in stablecoin use after support, the relation with continuous relative depth, and the difference in first-month adoption between shallow and more competitive stablecoin bridges. Because the persistence condition uses later capital, this event serves as a robustness and timing exercise.

Figure 3 follows events whose supported stablecoin is first used 7 to 119 days after persistent support. Stablecoin weak-leg capital rises during the week before first use, including relative to the

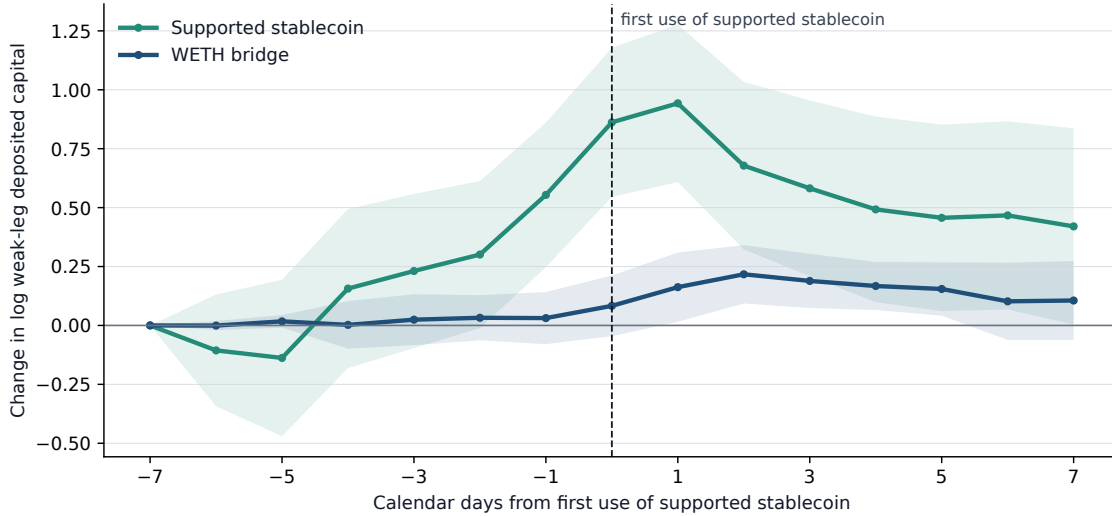
Table 27: Persistent stable-bridge support and route allocation

Outcome	First 30 days	Days 30–119
<i>Panel A. Changes around first persistent stable support</i>		
	+7.93*** (1.88)	+8.68*** (2.57)
Stable route share [pp]	+3.11** (1.53)	+6.94 (4.53)
Stable supported-value share [pp]	−0.20*** (0.04)	−0.39*** (0.06)
log(1 + native routes per active day)	−0.10** (0.04)	−0.29*** (0.05)
log(1 + total routes per active day)	−0.17 (0.11)	−0.61*** (0.19)
log(1 + native supported value per active day)	+0.06 (0.10)	−0.32* (0.18)
log(1 + total supported value per active day)		
Bridge events	865	818
<i>Panel B. Stable-bridge competitiveness relative to WETH</i>		
	+0.64*** (0.07)	+0.75*** (0.05)
Relative-depth slope [pp per +1 pp]		
Stable route share, depth < 0.1× WETH [%]	2.4 53.0	2.4 61.2
Stable route share, depth ≥ WETH [%]	(4.6) 69.9	(13.2) 70.7
Stable route share, depth ≥ 2× WETH [%]	(2.3)	(9.3)
Active pair-days	11,327	17,027
<i>Panel C. Adoption of supported stablecoin after persistent support</i>		
Outcome	First 30 days	First 120 days
Supported stablecoin used, depth < 0.1× WETH [%]	42.6	50.5
Supported stablecoin used, depth ≥ 0.1× WETH [%]	84.1	89.7
	+41.55*** (4.95)	+39.17*** (4.48)
Difference [pp]	+25.16*** (5.76)	+23.47*** (5.37)
Difference with controls [pp]	+35.76*** (11.89)	+40.26*** (11.66)
Difference, no stablecoin endpoint [pp]		

Note: The unit is a pair by route scope around the first persistent DAI, USDC, or USDT bridge. Support requires positive prior-calendar deposited capital on both legs in Uniswap v2 or SushiSwap v2 on at least 24 of the next 30 calendar days. Every pair has earlier WETH-mediated routes, and the event precedes its first observed stablecoin route. Panel A reports changes from the prior 30 days. Panel B compares the best stablecoin weak-leg capital with WETH weak-leg capital. Panel C records use of a stablecoin in the event’s support set within 30 or 120 days. Standard errors are two-way clustered by pair and date. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. Deposited capital is an equilibrium pool stock.

matched WETH bridge. The timing is consistent with providers deepening a bridge in anticipation of route demand, traders waiting for usable depth, or both.

Figure 3: Bridge capital around first use of the supported stablecoin



Note: The sample contains 246 pair-by-route-scope events whose first use of a stablecoin in the event’s support set occurs 7 to 119 calendar days after persistent support. Each line is the equal-event mean change from day -7 in $\log(1 + \text{deposited capital in USD})$ on the weaker leg. Stablecoin depth is the largest bottleneck among the supported stablecoins; WETH depth is the matched incumbent bottleneck. Capital is lagged to the prior calendar day. Shading gives 95% confidence intervals from standard errors two-way clustered by pair and adoption date.

Table 28 separates newly active pools from capital growth in pools already supporting the route. Newly active pools hold 7.5% of stablecoin weak-leg capital on the first-use day; 92.5% lies in pools active one week earlier. Most observed growth therefore comes from deepening pools that already exist, with a smaller pool-activation margin.

H.3 Relative-price risk and deposited bridge capital

Lower endpoint-vehicle relative-price risk is associated with deeper constant-product bridges at the measurement date and 30 days later. This relation is consistent with divergence risk influencing where liquidity providers place capital. Panel B of Table 29 gives the corresponding vehicle comparison. Stablecoin bridges have higher median risk than WETH bridges and carry the lower relative volatility in fewer than one third of comparable pair-months. Divergence risk helps locate bridge capital but leaves the aggregate rise of stablecoin intermediation unexplained.

Table 28: Pool activation and capital growth before stablecoin route use

Outcome	Stablecoin	WETH	Difference
Weak-leg deposited capital [log change]	+0.76*** (0.16)	+0.06 (0.07)	+0.70*** (0.17)
Active pool count [log change]	+0.04*** (0.01)	−0.01* (0.01)	+0.05*** (0.01)
Any newly active pool [pp]	+7.32*** (1.66)	+0.41 (0.41)	+6.91*** (1.72)
New-pool share of adoption capital [pp]	+7.47*** (1.70)	+0.41 (0.41)	+7.14*** (1.77)
Continuing-pool capital [log change]	+0.23*** (0.07)	+0.07* (0.04)	+0.16** (0.08)

Note: The sample contains the bridge events in Figure 3. Stablecoin and WETH outcomes use each vehicle’s weaker leg. A newly active pool has positive prior-calendar deposited capital on day zero and none on day -7; a continuing pool has positive capital on both dates. Each cell is an equal-event mean. Parentheses contain standard errors two-way clustered by pair and adoption date. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 29: Relative-price risk and deposited bridge capital

	(1) Depth at t	(2) Depth at t	(3) Depth at $t + 30$	(4) Depth at $t + 30$
<i>Panel A: Relative-price risk and bridge depth</i>				
Relative volatility [10 pp]	−0.117** (0.049)		−0.093*** (0.029)	
Daily divergence loss [1 bp]		−0.053** (0.021)		−0.044** (0.020)
Log prior vehicle-route count	+2.218*** (0.097)	+2.235*** (0.095)	+0.613*** (0.066)	+0.625*** (0.067)
Log initial bridge depth			+0.733*** (0.022)	+0.734*** (0.022)
Within R^2	0.322	0.321	0.702	0.702
Observations	58,447	58,447	58,447	58,447
Pair clusters	4,254	4,254	4,254	4,254
Date clusters	71	71	71	71
Pair × date fixed effects	Yes	Yes	Yes	Yes
Vehicle fixed effects	Yes	Yes	Yes	Yes
Two-way clustered s.e.	Pair, date	Pair, date	Pair, date	Pair, date
<i>Panel B: Native and stablecoin bridge risk</i>				
	Relative volatility		Daily divergence loss	
Native median	126.9%		5.60 bp	
Stablecoin median	148.4%		7.68 bp	
Stablecoin has lower risk [pair-months]	29.4%		28.6%	

Note: The unit is a pair, monthly date, and vehicle among WETH, DAI, USDC, and USDT. Each row requires at least ten routes for the pair during the preceding calendar month, at least 20 daily return observations during days -30 through -1, and positive deposited capital on both route legs for at least one native and one stablecoin bridge in the pair-month. Bridge depth is the smaller deposited-capital stock across the two legs in full-range Uniswap v2 and SushiSwap v2 pools, measured at the prior-calendar state. Relative volatility is the larger annualised endpoint–vehicle volatility across the two legs. Daily divergence loss applies the exact 50–50 constant-product expression to the same relative returns and takes the larger leg value. Columns 3 and 4 control for initial bridge depth; every column controls for prior vehicle-route activity and absorbs pair-by-date and vehicle effects. Standard errors are two-way clustered by pair and date. Asterisks *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. The estimates describe equilibrium liquidity allocation; fee income, provider hedges, and provider-level returns are unavailable.

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